Total, quantum, and classical measures of anticoherence for mixed spin states

Jérôme Denis,[1,2,*] Tara Lacaille,[2] John Martin,[2,†] and Eduardo Serrano-Ensástiga[3,2,‡]

[1] *Institute of Quantum Precision Measurement, State Key Laboratory of Radio Frequency Heterogeneous Integration, College of Physics and Optoelectronic Engineering, Shenzhen University, Shenzhen 518060, China*

[2] *Institut de Physique Nucléaire, Atomique et de Spectroscopie, CESAM, University of Liège B-4000 Liège, Belgium*

[3] *Departamento de Física, Universidad Autónoma Metropolitana-Iztapalapa, Av. Ferrocarril San Rafael Atlixco 186, C.P. 09310 Mexico City, Mexico*

(Dated: October 1, 2026)

Anticoherence is characterized by isotropic low-order spin moments. For pure states, it emerges from quantum superpositions and is relevant to direction-independent metrology and quantum reference-frame alignment. For mixed states such isotropy may also originate from classical statistical mixing. To formalize this difference, we introduce here an axiomatic framework for mixed-state t-anticoherence in the symmetric $N = 2j$ qubit representation, with $1 \leq t < N$. We distinguish *total* t-anticoherence, quantifying the reduced-state isotropy, from *quantum* t-anticoherence, defined relative to a chosen total measure, which quantifies the isotropy coming from entanglement between the t and $N - t$ qubits. This yields a *classical* contribution as their difference. For concave total measures, this classical contribution admits an ensemble/measurement interpretation analogous to a Koashi–Winter relation. We construct total anticoherence measures based on distances to or overlaps with the maximally mixed state and quantum counterparts via convex-roof extensions of pure-state anticoherence measures. For the trace distance and fidelity candidates, their general covariant-noise monotonicity remains open. We give analytic mixed-state examples, study robustness under particle loss for different types of states and analyze purity–anticoherence trade-offs, including certified thresholds for low spins.

I. INTRODUCTION

Spin states that lack any apparent spatial orientation to a certain degree, due to quantum superpositions, play an important role in quantum information, quantum metrology, and the theory of quantum reference frames. Such states are particularly relevant whenever no shared spatial reference frame is available, or when one seeks to suppress or average out directional information. This notion is captured by the concept of *anticoherent* (AC) spin states, originally introduced for pure states as those whose low-order spin moments are isotropic [1]. In particular, anticoherence can be viewed as the opposite of spin coherence: whereas coherent spin states are "most classical" in that they point maximally along some direction, AC states suppress directional moments (e.g. $\langle \mathbf{J} \rangle = 0$ at first order) by suitable quantum superpositions. AC states exhibit remarkable properties, ranging from connections to spherical designs and Majorana constellations to entanglement classification, robust geometric phases and optimal sensitivity bounds in direction-independent metrology [2–17].

Full rotational invariance and isotropy of low-order moments are different properties. In a fixed spin-j irreducible representation, the only state invariant under every rotation is the maximally mixed state (MMS). A t-AC state can retain multipoles of ranks greater than t, and can therefore encode orientation and be useful for estimating rotations. States that break rotational symmetry serve as reference-frame tokens [18–24]. The resource theory of SU(2) asymmetry formalizes this role using covariant operations [25–27]. In the present work, total t-anticoherence concerns only the information contained in multipoles of ranks at most t, equivalently in the t-qubit reduction; it does not measure the absence of all directional information.

Beyond their conceptual role, pure-state anticoherence has found concrete applications in several operational settings. In quantum metrology, highly AC states display optimal or near-optimal performance in fidelity-based estimation tasks that do not assume any preferred spatial direction, reflecting their uniform sensitivity to rotations and their close connection to spherical designs [4, 11, 16, 28]. More generally, AC states have been identified as optimal quantum rotosensors and as enabling multiparameter estimation of rotational (SO(3)) parameters (e.g. Euler angles) in suitable setups [8–11]. For mixed states, for instance in noisy rotational metrology, mixtures of AC states are also suitable for rotational metrological scenarios [28].

Anticoherence has also been investigated as a marker of *extreme quantumness*: maximally AC states can form orthonormal bases whose elements exhibit strong multipartite entanglement and pronounced nonclassical behavior [2]. Closely related ideas appear in reference-frame alignment protocols: states with strong low-order directional moments can serve as tokens of an absolute direction, whereas anticoherent states suppress such direc-

* jdenis@szu.edu.cn
† jmartin@uliege.be
‡ edensastiga@izt.uam.mx

tional bias and are instead well suited to isotropic estimation of rotations or frame misalignment [8]. These results motivate the experimental realization of anticoherent states [12, 29, 30] and similar states such as random and t-design states [31, 32].

While the theory of anticoherence for pure states is now well established, a systematic extension to mixed states is still lacking. For mixed states, isotropy of low-order moments may arise either from genuinely quantum correlations or from classical statistical mixing, such as averaging over unknown orientations. As a consequence, a single scalar quantity is generally insufficient to capture the physical content of anticoherence in the mixed-state regime, particularly from the viewpoint of quantum resources and asymmetry. This tension is already visible in existing quantifiers: several anticoherence measures have been proposed to quantify isotropy for arbitrary states [33], including distance-based constructions. Yet even the maximally mixed state (MMS), arguably the most classical mixed state, appears maximally anticoherent in the sense of low-order isotropy.

We distinguish total t-anticoherence, which quantifies reduced-state isotropy, from a quantum contribution based on entanglement across the $t\,|\,N-t$ split. The purity construction is a normalized convex roof of linear entropy of entanglement; the fidelity construction is normalized convex-roof extended negativity. These are established entanglement constructions. Their role here is to calibrate entanglement against the spin-multipole hierarchy and a total isotropy functional on the same scale. Their difference has a precise ensemble interpretation developed below, rather than a general interpretation as a state-preparation cost.

The two operational settings must be distinguished. Covariant noise is relevant to the total quantity and rotational asymmetry. The quantum contribution is instead constrained by local operations and classical communication (LOCC) between the groups of t and $N-t$ qubits. When the spin describes individually accessible constituent particles, this embedding has a direct operational realization; for an elementary spin, the constituent qubits are virtual and independent control of the two groups need not be physically available [6, 7, 34–36].

We formulate axioms for both types of measures, construct explicit examples based on distances to the MMS and on convex-roof extensions, and show that the total contribution always upper bounds the quantum contribution. Finally, we clarify the relation between our framework and existing tensor-based descriptions of angular structure, in particular the cumulative multipole measures defined and studied in [3, 37, 38].

The paper is organized as follows. Section II introduces mixed-state t-anticoherence and the axiomatic framework for total, quantum, and classical measures. Section III presents explicit constructions of these measures and their properties, including distance-based and cumulative-multipole formulations. Section IV provides explicit evaluations, and Section V discusses applications to representative mixed states, robustness under particle loss, and the trade-off between purity and the maximal achievable anticoherence order. We conclude in Sec. VI with some final comments.

II. AXIOMATIC FRAMEWORK FOR MIXED-STATE ANTICOHERENCE MEASURES

A. Mixed-state t-anticoherence

Let $\mathcal{H}_j$ denote the Hilbert space of a spin-j system, with standard angular-momentum basis $\{|j,m\rangle\}_{m=-j}^{j}$ (eigenstates of J^2 and J_z). Using the standard identification between $\mathcal{H}_j$ and the fully symmetric subspace of $N=2j$ qubits, any state $\rho\in\mathcal{D}(\mathcal{H}_j)$ defines a symmetric N-qubit state ρ_S. For $t\in\{1,\dots,N\}$ we denote by

$$\rho_t = \mathrm{Tr}_{N-t}(\rho_S) \tag{1}$$

the reduced state on t qubits, supported on the symmetric subspace of dimension $t+1$.

It is often convenient to characterize spin states in terms of irreducible tensor operators $\{T_{LM}\}$ [39][1]. Any spin-j state ρ can be expanded as

$$\rho = \frac{\mathbb{1}}{2j+1} + \sum_{L=1}^{2j}\sum_{M=-L}^{L} \rho_{LM} T_{LM}. \tag{2}$$

The coefficients $\rho_{LM}=\mathrm{Tr}(T_{LM}^{\dagger}\rho)$ are the multipole moments and encode angular structure at rank L. With the stated normalization, $T_{00}=\mathbb{1}/\sqrt{N+1}$ and $\rho_{00}=1/\sqrt{N+1}$.

Definition 1. *A spin state ρ, pure or mixed, is* t-anticoherent *(t-AC) if all its multipole moments up to rank t vanish, namely*

$$\rho_{LM}=0, \qquad 1\le L\le t, \quad -L\le M\le L. \tag{3}$$

Equivalently, all spin moments up to order t are isotropic.

Using the standard identification between $\mathcal{H}_j$ and the fully symmetric subspace of $N=2j$ qubits, this definition admits an equivalent reduced-state formulation. If ρ_S denotes the corresponding symmetric N-qubit state and ρ_t its reduced state on t qubits (1), then

$$\rho \text{ is } t\text{-AC} \quad\Leftrightarrow\quad \rho_t = \frac{\mathbb{1}_{t+1}}{t+1}. \tag{4}$$

Indeed, the partial trace map in the irreducible tensor basis reads [40]

$$\rho_t = \sum_{L=0}^{t}\sum_{M=-L}^{L} c_{N,t,L}\,\rho_{LM}\,T_{LM}^{(t)}, \tag{5}$$

[1] With the Hilbert–Schmidt normalization $\mathrm{Tr}(T_{LM}^{\dagger}T_{L'M'}) = \delta_{LL'}\delta_{MM'}$.

where

$$c_{N,t,L} = \frac{t!}{N!}\sqrt{\frac{(N-L)!(N+L+1)!}{(t-L)!(t+L+1)!}} \tag{6}$$

and $T^{(t)}_{LM}$ denotes the irreducible tensor operator acting on the spin-$t/2$ Hilbert space. Thus ρ_t depends only on the coefficients ρ_{LM} with $L \le t$, while all higher-rank components vanish under reduction. Consequently, ρ_t is maximally mixed if and only if all multipole moments of ρ up to rank t vanish.

Let us note two immediate consequences of Definition 1. First, t-anticoherence is monotone in the order: if ρ is t-AC, then it is t'-AC for every $t' \le t$. This follows either from the nesting of partial traces, $\rho_{t'} = \mathrm{Tr}_{t-t'}(\rho_t)$, together with $\rho_t = \mathbb{1}_{t+1}/(t+1)$, or equivalently from (5), since vanishing multipoles up to rank t implies vanishing multipoles up to rank t'. Second, t-anticoherence is stable under particle loss: for any $q \ge t$, if $\rho_q = \mathrm{Tr}_{N-q}(\rho_S)$, then $(\rho_q)_t = \mathrm{Tr}_{q-t}(\rho_q) = \mathrm{Tr}_{N-t}(\rho_S) = \rho_t$ by the associativity of the partial trace, so $\rho_t = \mathbb{1}_{t+1}/(t+1)$ implies $(\rho_q)_t = \mathbb{1}_{t+1}/(t+1)$, i.e., ρ_q is again t-AC.

Definition 1 reduces to the standard notion of t-anticoherence for pure states [41] given by Zimba [1]. For mixed states, however, isotropy up to order t can arise either from classical statistical mixing, genuine quantum correlations, or both; this motivates the distinction introduced in this work between total, quantum, and classical measures of anticoherence.

B. Total t-AC measure

A total t-AC measure quantifies isotropy of the t-qubit reduction independently of its origin. Throughout the whole manuscript, we assume $N \ge 2$ and $1 \le t < N$, so that the associated split $t \,|\, N-t$ is nontrivial. Definition 1 of the AC property also allows $t = N$.

Definition 2. *A function* $\mathcal{A}^T_t : \mathcal{D}(\mathcal{H}_j) \to [0,1]$ *is a* total t-AC measure *if it satisfies:*

(T1) $\mathcal{A}^T_t(\rho) = 1$ *iff* ρ *is t-AC;*

(T2) $\mathcal{A}^T_t(\rho) = 0$ *iff* ρ *is pure and coherent;*

(T3) $\mathcal{A}^T_t(U\rho U^\dagger) = \mathcal{A}^T_t(\rho)$ *for all* $U \in \mathrm{SU}(2)$*;*

(T4) Monotonicity under SU(2)*-covariant noise: For every* SU(2)*-covariant channel* $\Phi : \mathcal{D}(\mathcal{H}_j) \to \mathcal{D}(\mathcal{H}_j)$*, i.e. every completely positive trace-preserving map such that*

$$\Phi(U\rho U^\dagger) = U\,\Phi(\rho)\,U^\dagger \tag{7}$$

for all $U \in \mathrm{SU}(2)$ *and* $\rho \in \mathcal{D}(\mathcal{H}_j)$*, one has*

$$\mathcal{A}^T_t\big(\Phi(\rho)\big) \ge \mathcal{A}^T_t(\rho) \tag{8}$$

for all $\rho \in \mathcal{D}(\mathcal{H}_j)$*.*

Axioms (T3)–(T4) connect total t-AC to rotational asymmetry [25–27, 42, 43]. The free operations are SU(2)-covariant channels, and the unique invariant state in a fixed irreducible sector is $\rho_0 = \mathbb{1}_{2j+1}/(2j+1)$. Thus $1 - \mathcal{A}^T_t$ is a nonnegative but nonfaithful asymmetry monotone: it vanishes on t-AC states even when nonzero higher-rank multipoles prevent full rotational invariance. A large $\mathcal{A}^T_t$ means that the t-qubits marginal is close to isotropic according to the chosen functional. It does not by itself specify reference-frame alignment fidelity or rotational sensitivity.

The next lemma presents a basic structural property of SU(2)-covariant channels on a fixed spin sector, which will be used repeatedly in the construction of explicit total measures.

Lemma 1 ([44])**.** *Let* Φ *be an* SU(2)*-covariant channel on* $\mathcal{H}_j$*. Then* Φ *is unital, i.e.,* $\Phi(\mathbb{1}) = \mathbb{1}$*, and in particular the MMS* ρ_0 *is a fixed point of* Φ*.*

Lemma 1 explains why the isotropic MMS will naturally play the role of reference point in the distance- and purity-based construction of AC measures: any SU(2)-covariant channel fixes it.

Random global rotations with a conjugation-invariant probability density,

$$\rho \;\mapsto\; \int_{\mathrm{SU}(2)} p(U)\, U\rho U^\dagger \,\mathrm{d}U, \qquad p(VUV^\dagger) = p(U), \tag{9}$$

form a natural subclass of SU(2)-covariant channels and describe *classical* uncertainty about the laboratory orientation. They model the absence of a shared reference frame or deliberate orientation averaging [25]. However, the complete set of SU(2)-covariant channels is strictly larger: it contains all CPTP maps that commute with the rotation action, including irreversible dissipative processes [40, 45, 46] and measure-and-reprepare (entanglement-breaking) operations, provided they remain *direction insensitive*.

Since the operator space on a fixed spin-j irrep decomposes multiplicity-free into irreducible tensor sectors, covariance implies, by Schur's lemma, that Φ acts diagonally on irreducible tensor operators,

$$\Phi(T_{LM}) = f_L\, T_{LM} \qquad \forall\, L, M, \tag{10}$$

with f_L real numbers such that $|f_L| \le 1$; in particular, one always has $f_0 = 1$ (trace preservation), and complete positivity imposes nontrivial constraints on the admissible values of $(f_1, \ldots, f_{2j})$, see [47, 48] [49][2]. Therefore,

[2] For random global rotations of the form (9) with p depending only on the rotation angle, the coefficients f_L are the angular averages of the corresponding irreducible characters,

$$f_L = \frac{1}{2L+1}\int_{\mathrm{SU}(2)} p(U)\,\chi^{(L)}(U)\,\mathrm{d}U, \tag{11}$$

from the expansion (2), one has

$$\Phi(\rho) = \frac{\mathbb{1}}{2j+1} + \sum_{L=1}^{2j} \sum_{M=-L}^{L} f_L \, \rho_{LM} \, T_{LM}. \tag{12}$$

Thus a covariant channel acts with the same factor on every M at fixed L, while the factors f_L for different ranks are jointly constrained by complete positivity. They cannot be chosen independently from the intervals $[-1, 1]$.

C. Quantum t-anticoherence measure

While total t-anticoherence quantifies isotropy independently of its origin, it is useful to single out the contribution stemming from genuinely quantum correlations. For mixed states, this leads to the notion of *quantum* t-anticoherence measures, which must behave as resource monotones and remain non-increasing under probabilistic state preparation.

Given a total t-anticoherence measure $\mathcal{A}_t^T$ according to Definition 2, we define a corresponding notion of quantum t-anticoherence as follows.

Definition 3. *A function $\mathcal{A}_t^Q : \mathcal{D}(\mathcal{H}_j) \to [0,1]$ is called a* quantum t-anticoherence measure associated with $\mathcal{A}_t^T$ *if it satisfies:*

(Q1) Vanishing on separable states: $\mathcal{A}_t^Q(\rho) = 0$ *for every ρ corresponding to a* fully separable *multiqubit symmetric state ρ_S*[3].

(Q2) Pure-state consistency with the chosen total measure[4]: *for all pure states $|\psi\rangle$,*

$$\mathcal{A}_t^Q(|\psi\rangle\langle\psi|) = \mathcal{A}_t^T(|\psi\rangle\langle\psi|). \tag{14}$$

(Q3) SU(2) invariance: $\forall\, U \in \mathrm{SU}(2)$

$$\mathcal{A}_t^Q(U\rho U^\dagger) = \mathcal{A}_t^Q(\rho). \tag{15}$$

(Q4) Convexity: $\mathcal{A}_t^Q(\sum_i p_i\rho_i) \le \sum_i p_i \mathcal{A}_t^Q(\rho_i)$.

(Q5) Average monotonicity under symmetric-sector LOCC: *For any state ρ and every LOCC protocol across the bipartition $t\,|\,N{-}t$ in the symmetric N-qubit embedding, whose final normalized outcomes are supported on the symmetric subspace,*

$$\mathcal{A}_t^Q(\rho) \ge \sum_k p_k \mathcal{A}_t^Q(\rho_k), \tag{16}$$

where ρ_k are the corresponding spin-state representatives, obtained with probabilities p_k.

(Q6) Ordering: $\forall\, \rho \in \mathcal{D}(\mathcal{H}_j)$

$$\mathcal{A}_t^Q(\rho) \le \mathcal{A}_t^T(\rho). \tag{17}$$

In the symmetric subspace, separability across a nontrivial bipartition is equivalent to full separability. A symmetric state outside the convex hull of spin-coherent states is therefore genuinely multipartite entangled [50]. The quantum quantities constructed below quantify this entanglement with a normalization adapted to order-t isotropy. Convexity (Q4) and rotational invariance (Q3) imply that they cannot increase under random global rotations.

Axiom (Q5) is an entanglement requirement on the symmetric-qubit embedding. It describes physically available operations on the two groups only when that access to the constituent particles is part of the experimental setting. Particle loss changes N and therefore requires a separate comparison of measures on different spin sectors, beyond these fixed-N axioms.

D. Classical t-anticoherence contribution

The difference between total and quantum t-AC quantifies the contribution of classical mixing to isotropy.

Definition 4. *Given a total measure $\mathcal{A}_t^T$ and its associated quantum measure $\mathcal{A}_t^Q$, we define the* classical t-AC contribution $\mathcal{A}_t^C : \mathcal{D}(\mathcal{H}_j) \to [0,1]$ *by*

$$\mathcal{A}_t^C(\rho) \equiv \mathcal{A}_t^T(\rho) - \mathcal{A}_t^Q(\rho). \tag{18}$$

Axiom (Q6) guarantees nonnegativity, and pure-state consistency guarantees that $\mathcal{A}_t^C$ vanishes on pure states.

An ensemble interpretation of this difference is developed in Sec. III B, after introducing the quantum-measure construction.

If the total functional $\mathcal{A}_t^T$ is concave, then $\mathcal{A}_t^C$ is concave because $\mathcal{A}_t^Q$ is convex. Together with rotational invariance, this implies that $\mathcal{A}_t^C$ cannot decrease under any random mixture of global rotations; the rotation distribution need not be conjugation invariant.

The separation $\mathcal{A}_t^T = \mathcal{A}_t^Q + \mathcal{A}_t^C$ makes explicit that a state may display large (or even maximal) *total* t-anticoherence while having no *quantum* contribution, i.e.

---

where $\chi^{(L)}$ is the character of the spin-L representation. In particular, $f_0 = 1$ (trace preservation), and for $L \ge 1$ the values f_L quantify how strongly the random rotations damp the rank-L tensor components.

[3] Equivalently, $\mathcal{A}_t^Q(\rho) = 0$ for every symmetric state that is separable across the bipartition $t\,|\,N{-}t$, since in the symmetric subspace bipartite separability implies full separability. Such symmetric states are of the form

$$\rho_S = \int \big(|n\rangle\langle n|\big)^{\otimes N} \, \mathrm{d}\nu(n) \tag{13}$$

with ν a normalized nonnegative measure, which can have discrete support.

[4] In particular, if $\mathcal{A}_t^T$ coincides on pure states with a previously introduced pure-state anticoherence measure $\mathcal{A}_t$ in [33], then $\mathcal{A}_t^Q$ coincides with $\mathcal{A}_t$ on pure states as well.

$\mathcal{A}_t^Q = 0$ and $\mathcal{A}_t^C = \mathcal{A}_t^T$. The most extreme example is the MMS ρ_0, which is t-AC for every t since all multipole moments vanish, yet is a separable symmetric state and is therefore assigned $\mathcal{A}_t^Q(\rho_0) = 0$ (hence $\mathcal{A}_t^C(\rho_0) = 1$) for any possible quantum t-AC measure.[5]

LOCC monotonicity is not required of this classical contribution. Classical communication can create correlations, which is why standard classical-correlation measures require monotonicity under local operations without communication [51]. Here, shared uniform rotations take a coherent state to ρ_0, raising $\mathcal{A}_t^C$ from zero to one when the rotation record is discarded; local resetting to a common coherent direction reverses this change. Both are LOCC channels with symmetric outputs. Thus neither direction of LOCC monotonicity holds.

Both endpoint states are classical in the usual spin sense: they belong to the convex hull of coherent states [52]. Their different $\mathcal{A}_t^C$ values quantify different amounts of isotropy produced by mixing. This distinction is analogous to separating classical and quantum contributions to uncertainty [53].

A second example is provided by the equal incoherent mixture of two opposite spin-coherent states along a fixed axis,

$$\rho = \tfrac{1}{2}\,|j,j\rangle\langle j,j| + \tfrac{1}{2}\,|j,-j\rangle\langle j,-j|\,, \tag{19}$$

which has $\langle \mathbf{J}\rangle = 0$ (and thus is perfectly isotropic at first order $t = 1$) purely by classical averaging of two antipodal directions. In particular, ρ has $\mathcal{A}_1^C(\rho) = \mathcal{A}_1^T(\rho) = 1$ for any 1-AC measure.

III. CONSTRUCTIONS OF AC MEASURES

A. Total AC measures: General constructions

We begin by constructing several measures of total anticoherence.

1. *Purity-based construction*

A particularly simple example of a total t-anticoherence measure is obtained from the purity of the reduced symmetric t-particle state. We define

$$\mathcal{A}_t^{T,\mathcal{P}}(\rho) \equiv \frac{t+1}{t}\Big(1 - \mathrm{Tr}(\rho_t^2)\Big). \tag{20}$$

This expression coincides with the standard purity-based measure of t-anticoherence $\mathcal{A}_t^R$ for pure states [33] and provides a direct extension to mixed states.

[5] The separability of ρ_0 follows from the uniform coherent-state mixture $\rho_0 = \int(|n\rangle\langle n|)^{\otimes N}\,\mathrm{d}n/(4\pi)$ in the symmetric sector. This state is the normalized projector onto that sector, rather than the identity on the full 2^N-dimensional qubit space.

Proposition 1 (Proof in Appendix A 1)**.** *The quantity $\mathcal{A}_t^{T,\mathcal{P}}(\rho)$ satisfies axioms (T1)–(T4) and therefore defines a measure of total t-anticoherence.*

2. *Distance-based construction*

Distance-based constructions provide another natural class of total t-anticoherence candidates. Let $d(\cdot,\cdot)$ be a unitarily invariant operator distance on the state space of the symmetric t-qubit sector. We normalize the distance from the MMS $\rho_0^{(t)} \equiv \mathbb{1}_{t+1}/(t+1)$, by its largest possible value,

$$K_t = \max_{\sigma\in\mathcal{D}(\mathcal{H}_{t/2})} d(\sigma, \rho_0^{(t)}). \tag{21}$$

This leads to the normalized distance-based total anticoherence candidate

$$\mathcal{A}_t^{T,d}(\rho) \equiv 1 - \frac{d(\rho_t, \rho_0^{(t)})}{K_t}. \tag{22}$$

With this convention, states whose t-body reduction equals the MMS have maximal total t-anticoherence, while states farthest from the MMS have minimal value. For the standard distances considered below, whose maximal distance from $\rho_0^{(t)}$ is attained exactly on pure states, this construction satisfies axioms (T1), (T2), and (T3). If the maximum is attained only when ρ_t is pure, then, because ρ_t is a marginal of a symmetric N-qubit state, this implies that the original state is a pure spin-coherent state. Only (T4) must be proved separately for all SU(2)-covariant channels.

In particular, for a fixed integer $p \geq 1$, we have the distances

$$d_p(\rho,\sigma) = ||\rho - \sigma||_p, \tag{23}$$

where $\rho,\sigma \in \mathcal{D}(\mathcal{H}_j)$, based on the Schatten p-norm

$$||A||_p = \left(\sum_{i=1}^{r} \lambda_i^p\right)^{1/p}, \tag{24}$$

where λ_i and r are the singular values and the rank of A, respectively.

Proposition 2 (Proof in Appendix A 2)**.** *The quantity $\mathcal{A}_t^{T,d}(\rho)$ for the distance based on the Schatten 2-norm (associated with the Hilbert-Schmidt distance) satisfies axioms (T1)-(T4) and therefore defines a measure of total t-anticoherence.*

For the construction based on the Hilbert-Schmidt distance,

$$K_t^{\mathrm{HS}} = \sqrt{\frac{t}{t+1}}, \qquad \mathcal{A}_t^{T,\mathrm{HS}} = 1 - \sqrt{1 - \mathcal{A}_t^{T,\mathcal{P}}}. \tag{25}$$

It therefore orders states in the same way as the purity total, although a nonlinear transformation does not generally commute with taking a convex roof. For trace distance, $K_t^{\mathrm{tr}} = 2t/(t+1)$. Its total functional obeys (T1)–(T3), while (T4) is proved below for $t = 1$ and for random rotations and remains open for general covariant channels at higher orders.

3. *Fidelity-based construction*

Another construction is obtained from the Uhlmann–Jozsa fidelity between the reduced state ρ_t and the MMS $\rho_0^{(t)} = \mathbb{1}_{t+1}/(t+1)$ on the symmetric t-qubit subspace:

$$F(\rho_t, \rho_0^{(t)}) \equiv \left(\mathrm{Tr}\sqrt{\sqrt{\rho_t}\rho_0^{(t)}\sqrt{\rho_t}}\right)^2 = \frac{(\mathrm{Tr}\sqrt{\rho_t})^2}{t+1}. \quad (26)$$

We choose an affine rescaling and define

$$\mathcal{A}_t^{T,F}(\rho) \equiv \frac{(t+1)F - 1}{t} = \frac{(\mathrm{Tr}\sqrt{\rho_t})^2 - 1}{t}. \quad (27)$$

Equivalently,

$$\mathcal{A}_t^{T,F}(\rho) = \frac{(t+1)\left[1 - \frac{1}{2}D_{\mathrm{B}}\left(\rho_t, \rho_0^{(t)}\right)^2\right]^2 - 1}{t} \quad (28)$$

in terms of the Bures distance

$$D_{\mathrm{B}}(\sigma, \tau) = \sqrt{2\left(1 - \sqrt{F(\sigma, \tau)}\right)}. \quad (29)$$

The normalization is such that $\mathcal{A}_t^{T,F} = 1$ for t-AC states and $\mathcal{A}_t^{T,F} = 0$ for spin-coherent states. Indeed, $F(\rho_t, \rho_0^{(t)}) = 1$ iff $\rho_t = \rho_0^{(t)}$, whereas the minimum value $F(\rho_t, \rho_0^{(t)}) = 1/(t+1)$ is reached iff ρ_t is pure, which in the symmetric setting implies that ρ is a pure spin-coherent state. The SU(2) invariance follows from the unitary invariance of the fidelity, $F(U\rho_t U^\dagger, U\sigma U^\dagger) = F(\rho_t, \sigma)$, together with $U\rho_0^{(t)}U^\dagger = \rho_0^{(t)}$. Thus $\mathcal{A}_t^{T,F}$ satisfies axioms (T1)–(T3). Concerning axiom (T4), some care is needed. Although a SU(2)-covariant channel Φ on the full spin-j space induces a linear transformation on the reduced state ρ_t, this induced reduced map is not, in general, a completely positive trace-preserving channel acting on the whole state space $\mathcal{D}(\mathcal{H}_{t/2})$. Therefore one cannot simply invoke monotonicity of the Uhlmann fidelity for an induced channel fixing $\rho_0^{(t)}$. Numerical evidence suggests that $\mathcal{A}_t^{T,F}$ is nevertheless non-decreasing under SU(2)-covariant noise, but we leave a proof of axiom (T4) for future work.

The following cases of (T4) can be proved directly. At $t = 1$, set $r = |\langle \mathbf{J} \rangle|/j$. Then

$$\begin{aligned} \mathcal{A}_1^{T,\mathcal{P}} &= 1 - r^2, \\ \mathcal{A}_1^{T,F} &= \sqrt{1 - r^2}, \\ \mathcal{A}_1^{T,\mathrm{HS}} &= \mathcal{A}_1^{T,\mathrm{tr}} = 1 - r. \end{aligned} \quad (30)$$

Covariance sends r to $|f_1|r \le r$, proving (T4) for both candidates at first order. At every proper order, concavity and rotational invariance of a total anticoherence measure prove its nondecrease under any random-rotation channel (the channel is covariant when the rotation distribution is central). For fidelity, the required concavity is established in Appendix A 7; for trace distance it follows from convexity of the norm.

Conjecture 1. *For every $2 \le t < N$, the trace-distance and fidelity total candidates are nondecreasing under all SU(2)-covariant channels on the spin-$N/2$ sector.*

Numerical tests described in Appendix C support this conjecture; they do not establish it.

We conclude this section with a general result, which will prove useful later on the relation between the Uhlmann fidelity and entanglement. Consider a quantum state $|\psi\rangle$ defined on a composite Hilbert space $\mathcal{H}_{AB}$. Then, a measure of entanglement of the state $|\psi\rangle$ on the bipartition $A|B$ is given by the entanglement negativity [54]

$$\mathcal{N}_A = \sum_{i>j=1}^{d_A} \alpha_i \alpha_j \quad (31)$$

where the α_i are the Schmidt coefficients (padded with zeros if necessary) of $|\psi\rangle$ with respect to the split $A|B$. From this definition, we have the following result.

Proposition 3 (Proof in Appendix A 3)**.** *For any composite system AB, a pure state $|\psi\rangle \in \mathcal{H}_{AB}$ verifies*

$$F(\rho_A, \rho_0) = \frac{1 + 2\mathcal{N}_A(|\psi\rangle)}{d_A}, \quad (32)$$

where ρ_A is the reduced state defined on A obtained by tracing out the subsystem B and $\rho_0 = \mathbb{1}_A/d_A$ is the MMS in $\mathcal{H}_A$.

This result gives a direct correspondence between the fidelity-based AC measures (27) of a pure state and its entanglement with respect to the bipartition $t|N-t$.

B. Quantum AC measures: general constructions via convex-roof

We now construct quantum t-AC measures by convex-roof extension. We use $\mathcal{A}_t^X(\psi) = \mathcal{A}_t^{T,X}(|\psi\rangle\langle\psi|)$ for the pure-state restriction of construction $X \in \{\mathcal{P}, \mathrm{HS}, \mathrm{tr}, F\}$.

1. *Purity-based construction*

For pure symmetric states, the quantity $1 - \mathrm{Tr}(\rho_t^2)$ coincides with the linear entropy of entanglement across

the bipartition $t\,|\,N{-}t$. This motivates defining a quantum t-AC measure for mixed states via the convex-roof construction,

$$\mathcal{A}_t^{Q,\mathcal{P}}(\rho) = \min_{\{p_i,|\psi_i\rangle\}} \sum_i p_i\, \mathcal{A}_t^{\mathcal{P}}(|\psi_i\rangle), \tag{33}$$

where the minimization runs over all pure-state decompositions of ρ.

Proposition 4 (Proof in Appendix A 4). *The quantity $\mathcal{A}_t^{Q,\mathcal{P}}(\rho)$ satisfies axioms (Q1)–(Q6) and therefore defines a quantum measure of t-AC associated with the total purity-based measure $\mathcal{A}_t^{T,\mathcal{P}}$.*

Proposition 5 (Proof in Appendix A 5). *For all mixed states $\rho \in \mathcal{D}(\mathcal{H}_j)$ and all $t \in \{1, 2, \ldots, N-1\}$,*

$$\mathcal{A}_t^{Q,\mathcal{P}}(\rho) = \frac{(t+1)(N-t)}{t(N+1-t)}\, \mathcal{A}_{N-t}^{Q,\mathcal{P}}(\rho). \tag{34}$$

The fidelity construction obeys the analogous relation

$$\mathcal{A}_t^{Q,F}(\rho) = \frac{N-t}{t} \mathcal{A}_{N-t}^{Q,F}(\rho). \tag{35}$$

2. Based on convex distances

We will use a distance d that is convex in its first argument. That is, for all $\rho_1, \rho_2, \sigma \in \mathcal{D}(\mathcal{H}_j)$ and all $\lambda \in [0,1]$, it satisfies

$$d(\lambda\rho_1 + (1-\lambda)\rho_2, \sigma) \le \lambda d(\rho_1,\sigma) + (1-\lambda) d(\rho_2,\sigma). \tag{36}$$

Given such a convex distance d, we extend the pure-state quantum t-AC measure to mixed states via the convex-roof construction:

$$\mathcal{A}_t^{Q,d}(\rho) = \min_{\{p_i,|\psi_i\rangle\}} \sum_i p_i \mathcal{A}_t^{d}(|\psi_i\rangle), \tag{37}$$

where the minimization is taken over all pure-state decompositions of ρ, and $\mathcal{A}_t^d$ denotes the pure-state t-AC measure induced by the distance d. When the pure-state functional $\mathcal{A}_t^d(|\psi\rangle)$ is an entanglement monotone across the bipartition $t\,|\,N{-}t$, its convex-roof extension provides a valid quantum t-AC measure associated with the corresponding total distance-based quantity.

Proposition 6 (Proof in Appendix A 6). *For any distance d whose associated total quantity $\mathcal{A}_t^{T,d}$ satisfies the total-measure axioms, and for which the pure-state functional $\mathcal{A}_t^d(|\psi\rangle)$ is an entanglement monotone across the $t\,|\,N{-}t$ bipartition, the convex-roof extension $\mathcal{A}_t^{Q,d}$ satisfies axioms (Q1)–(Q5). If, in addition, $\mathcal{A}_t^{T,d}$ is concave on $\mathcal{D}(\mathcal{H}_j)$, then the ordering axiom (Q6) also holds. Under these assumptions, $\mathcal{A}_t^{Q,d}$ defines a quantum measure of t-AC associated with the corresponding total distance-based measure $\mathcal{A}_t^{T,d}$.*

3. Fidelity-based construction

We now define the quantum measure associated with the total fidelity-based quantity (28). The key point is that, on pure states, the fidelity appearing in (27) is directly related to the negativity across the bipartition $t\,|\,N{-}t$. Applying Proposition 3 with $d_A = t+1$ gives, for every pure symmetric state $|\psi\rangle$,

$$F\big(\rho_t, \rho_0^{(t)}\big) = \frac{1 + 2\mathcal{N}_t(|\psi\rangle)}{t+1}. \tag{38}$$

Consequently,

$$\mathcal{A}_t^{T,F}(|\psi\rangle) = \frac{2}{t}\, \mathcal{N}_t(|\psi\rangle). \tag{39}$$

This suggests defining the associated quantum t-AC measure by the normalized convex-roof negativity

$$\mathcal{A}_t^{Q,F}(\rho) = \frac{2}{t}\, \mathcal{N}_t^{\mathrm{CR}}(\rho), \tag{40}$$

where

$$\mathcal{N}_t^{\mathrm{CR}}(\rho) = \min_{\{p_i,|\psi_i\rangle\}} \sum_i p_i\, \mathcal{N}_t(|\psi_i\rangle) \tag{41}$$

is the convex-roof extension of the negativity across the $t\,|\,N{-}t$ bipartition [55].

Proposition 7 (Proof in Appendix A 7). *The fidelity-based total candidate $\mathcal{A}_t^{T,F}$ and the quantum quantity $\mathcal{A}_t^{Q,F}$ defined in Eqs. (28) and (40) satisfy the following properties: The total candidate $\mathcal{A}_t^{T,F}$ satisfies axioms (T1)–(T3); it satisfies axiom (T4) at $t = 1$ and under covariant random rotations at every proper order, while the general higher-order statement is Conjecture 1. The quantum quantity $\mathcal{A}_t^{Q,F}$ satisfies axioms (Q1), (Q3)–(Q5), coincides with $\mathcal{A}_t^{T,F}$ on pure states as required by (Q2), and obeys the ordering axiom (Q6) with respect to this total candidate. In particular,*

$$\mathcal{A}_t^{Q,F}(\rho) \le \mathcal{A}_t^{T,F}(\rho) \qquad \forall\, \rho \in \mathcal{D}(\mathcal{H}_j). \tag{42}$$

Final remarks. The previous propositions show that, for the purity-based and Hilbert–Schmidt distance-based constructions, the corresponding quantum measures satisfy axioms (Q1)–(Q6) relative to the chosen total measure. For the fidelity-based construction, the quantum quantity satisfies the quantum-side axioms and the ordering $\mathcal{A}_t^{Q,F} \le \mathcal{A}_t^{T,F}$ relative to the total candidate (28), so the corresponding classical contribution $\mathcal{A}_t^{C,F} = \mathcal{A}_t^{T,F} - \mathcal{A}_t^{Q,F}$ is nonnegative. However, since axiom (T4) for $\mathcal{A}_t^{T,F}$ is supported here at general higher orders only by numerical evidence, the fidelity-based pair should be regarded as a well-motivated candidate total–quantum pair until a proof of total monotonicity under all SU(2)-covariant channels is obtained.

C. Classical AC measures: general constructions via convex-roof

Consider a source that prepares $|\psi_i\rangle$ with probability p_i. Knowing which state was prepared gives the average anticoherence $\sum_i p_i \mathcal{A}_t^T(\psi_i)$. Without this information, the state is $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$: its directional moments can cancel through mixing. For a concave total functional, the mixture has at least this average anticoherence.

The convex roof minimizes that average over all pure-state decompositions of ρ. Hence the classical contribution is the largest possible increase:

$$\mathcal{A}_t^C(\rho) = \max_{\{p_i,|\psi_i\rangle\}} \left[\mathcal{A}_t^T(\rho) - \sum_i p_i \mathcal{A}_t^T(\psi_i) \right]. \tag{43}$$

The maximization makes this a property of ρ, independent of its actual preparation. This interpretation relies on the convex-roof construction.

An equivalent measurement interpretation makes the role of this information explicit. Let A contain the t retained qubits and B the remaining ones. Introduce an auxiliary system E so that the joint state of ABE is pure and its AB marginal is ρ. Whenever $\mathcal{A}_t^T(\rho) = f_t(\rho_A)$ with f_t concave, which is the case for the purity, Hilbert-Schmidt, trace and fidelity total AC measures,

$$\mathcal{A}_t^C(\rho) = \max_{\{M_i\} \text{ on } E} \left[f_t(\rho_A) - \sum_i p_i f_t(\rho_{A|i}) \right], \tag{44}$$

where $\{M_i\}$ is a POVM, and $p_i, \rho_{A|i}$ are the outcome probabilities and conditional states of A. Rank-one measurements on E generate all pure-state ensembles of ρ; concavity ensures that refining a measurement cannot reduce the expression. Thus $\mathcal{A}_t^C$ is the maximal average reduction in marginal isotropy obtained by learning an outcome on E. The relation (44) is the analogue of the Koashi-Winter relation between classical correlations of A with E, entanglement of formation and von Neumann entropy [56]. For our functionals, it gives a generalized entropy-reduction interpretation, without assuming all monotonicity properties of the entropy-based correlation measure.

For the purity AC measure, Eq. (43) admits a direct variance interpretation. Set $\sigma_i = (|\psi_i\rangle\langle\psi_i|)_t$ and $c_t = (t+1)/t$. Then

$$\begin{aligned} \mathcal{A}_t^{C,\mathcal{P}}(\rho) &= c_t \max_{\{p_i,|\psi_i\rangle\}} \sum_i p_i \|\sigma_i - \rho_t\|_2^2 \qquad (45) \\ &= \frac{c_t}{2} \max_{\{p_i,|\psi_i\rangle\}} \sum_{i,k} p_i p_k \|\sigma_i - \sigma_k\|_2^2. \end{aligned}$$

Expanding in irreducible tensors gives

$$\mathcal{A}_t^{C,\mathcal{P}}(\rho) = \frac{t+1}{t} \max_{\{p_i,|\psi_i\rangle\}} \sum_{L=1}^{t} \sum_{M=-L}^{L} c_{N,t,L}^2 \operatorname{Var}_p(\rho_{LM}^{(i)}), \tag{46}$$

where $c_{N,t,L}$ is the partial-trace coefficient defined in (6), $\rho_{LM}^{(i)}$ is the state multipole of $|\psi_i\rangle$ and

$$\operatorname{Var}_p(\rho_{LM}^{(i)}) = \sum_i p_i |\rho_{LM}^{(i)}|^2 - \left| \sum_i p_i \rho_{LM}^{(i)} \right|^2. \tag{47}$$

One common ensemble maximizes the sum; the individual multipole variances cannot be optimized independently. The classical term thus quantifies the largest compatible dispersion of reduced states, or equivalently of their low-order multipoles.

D. Cumulative-multipole constructions

Cumulative multipole measures introduced and studied in [3, 37, 38, 57, 58] provide a tensor-based characterization of the angular structure of spin states. Related work connects these multipoles with higher-order polarization coherences and quantum polarization structure [57, 58]. In particular, they recover the standard distinction between first-order (Stokes-vector) polarization and higher-order, "hidden" angular structure by varying the cutoff rank: the case $t = 1$ isolates first-order directional content, while $t \geq 2$ probes the presence (or suppression) of higher-order multipoles. In our setting, this same hierarchy fits naturally with mixed-state total anticoherence, and could be further refined by separating, for each t, the total isotropy into quantum and classical contributions.

From the irreducible tensor-operator expansion (2), define the rank-L multipole weight and its cumulative version by

$$a_L(\rho) = \sum_{M=-L}^{L} |\rho_{LM}|^2, \qquad C_{\leq t}(\rho) = \sum_{L=1}^{t} a_L(\rho). \tag{48}$$

One has $C_{\leq t}(\rho) = 0$ iff ρ is t-AC, i.e. all multipoles up to rank t vanish. Moreover, the multipole weights $\sum_{M=-L}^{L} |\rho_{LM}|^2$ can be expressed as linear combinations of the reduced-state purities $\operatorname{Tr}(\rho_k^2)$ (see [11]), which makes explicit the link with the purity-based constructions.

1. *Total cumulative-multipole AC*

For $1 \leq t \leq 2j-1$, a natural *total* t-anticoherence measure is obtained by an affine decreasing rescaling of $C_{\leq t}$ that vanishes on pure coherent states. For spin-coherent states $\rho_{\text{coh}} = |\theta,\phi\rangle\langle\theta,\phi|$, one finds (independently of θ, ϕ)

$$C_{\leq t}(\rho_{\text{coh}}) = \frac{2j}{2j+1} - \frac{((2j)!)^2}{(2j-t-1)!(2j+t+1)!}. \tag{49}$$

In this subsection, we therefore restrict ourselves to $1 \leq t \leq 2j-1$ and define

$$\mathcal{A}_t^{T,\text{cm}}(\rho) = 1 - \frac{C_{\leq t}(\rho)}{C_{\leq t}(\rho_{\text{coh}})}. \tag{50}$$

Proposition 8 (Proof in Appendix A 8). *For $1 \le t \le 2j-1$, the quantity $\mathcal{A}_t^{T,\mathrm{cm}}$ satisfies axioms (T1)–(T4), and therefore defines a valid measure of total t-anticoherence.*

2. *Limitations of the cumulative-multipole convex roof*

A natural attempt at defining a corresponding *quantum* cumulative-multipole measure is to take the convex roof of the pure-state functional:

$$\mathcal{A}_t^{Q,\mathrm{cm}}(\rho) = \min_{\{p_i,|\psi_i\rangle\}} \sum_i p_i \, \mathcal{A}_t^{T,\mathrm{cm}}(|\psi_i\rangle\langle\psi_i|). \tag{51}$$

This convex roof satisfies (Q1)–(Q4) and (Q6), but a valid construction for all spins and orders would also require LOCC monotonicity of its pure-state restriction. The following counterexample at $N = 4, t = 2$ excludes such a general claim. In fact, it excludes every extension satisfying both pure-state consistency (Q2) and (Q5), independently of how the extension is constructed.

Consider the spin-2 pure states

$$|\psi\rangle = \frac{1}{\sqrt{3}}\Big(|2,2\rangle + |2,0\rangle + |2,-2\rangle\Big), \tag{52}$$

$$|\phi\rangle = \frac{1}{\sqrt{6}}\Big(-2|2,2\rangle - |2,0\rangle + |2,-2\rangle\Big). \tag{53}$$

Across the bipartition 2|2, their Schmidt spectra are

$$\lambda(\psi) = \left(\frac{7}{18} + \frac{\sqrt{6}}{9}, \frac{2}{9}, \frac{7}{18} - \frac{\sqrt{6}}{9}\right), \tag{54}$$

$$\lambda(\phi) = \left(\frac{4}{9} + \frac{\sqrt{87}}{36}, \frac{4}{9} - \frac{\sqrt{87}}{36}, \frac{1}{9}\right). \tag{55}$$

These satisfy $\lambda(\psi) \prec \lambda(\phi)$. Hence, by Nielsen's theorem [59], there exists a deterministic LOCC transformation $|\psi\rangle \longrightarrow |\phi\rangle$ across the split 2|2.

We now evaluate the cumulative multipole measure directly from Eq. (50) for $t = 2$ and obtain

$$\mathcal{A}_2^{T,\mathrm{cm}}(|\psi\rangle\langle\psi|) = \frac{7}{12}, \qquad \mathcal{A}_2^{T,\mathrm{cm}}(|\phi\rangle\langle\phi|) = \frac{31}{48}. \tag{56}$$

Since both states are pure, their total AC measure is equal to their quantum AC measure, and thus

$$\mathcal{A}_2^{Q,\mathrm{cm}}(|\psi\rangle\langle\psi|) = \frac{7}{12} < \frac{31}{48} = \mathcal{A}_2^{Q,\mathrm{cm}}(|\phi\rangle\langle\phi|), \tag{57}$$

although $|\psi\rangle \to |\phi\rangle$ is achievable by deterministic LOCC. This violates (Q5) at $N = 4, t = 2$ under the stated convention that initial and final states, rather than all intermediate states, must be symmetric.

Failure at this pair does not imply failure at every $t \ge 2$. For example, for $N = 3, t = 2$ the pure-state identity

$$\mathcal{A}_2^{T,\mathrm{cm}}(\psi) = \frac{9}{14}\mathcal{A}_1^{\mathcal{P}}(\psi) \tag{58}$$

implies that its convex roof is a valid entanglement monotone on the 2|1 split. To see the identity, let Θ be spin time reversal. For a pure half-integer-spin state, $\langle\psi|\Theta\psi\rangle = 0$. Since Θ changes the sign of odd-rank multipoles, $a_0 + a_2 = 1/2$ for $N = 3$, hence $a_2 = 1/4$. Also $a_1 = 9r^2/20$ and $C_{\le 2}(\rho_{\mathrm{coh}}) = 7/10$, giving Eq. (58). At $t = 1$ there is a general identity for every spin, as follows.

Proposition 9 (Proof in Appendix A 9). *For $t = 1$, the normalized total cumulative-multipole and purity constructions coincide on every state,*

$$\mathcal{A}_1^{T,\mathrm{cm}}(\rho) = \mathcal{A}_1^{T,\mathcal{P}}(\rho). \tag{59}$$

Consequently, their convex-roof quantum extensions also coincide:

$$\mathcal{A}_1^{Q,\mathrm{cm}}(\rho) = \mathcal{A}_1^{Q,\mathcal{P}}(\rho) \qquad \forall\, \rho \in \mathcal{D}(\mathcal{H}_j), \tag{60}$$

with the normalizations chosen above. In particular, $\mathcal{A}_1^{Q,\mathrm{cm}}$ satisfies axiom (Q5).

IV. EVALUATION OF MEASURES

Example 1. Rank-2 mixed states for $j = 2$

As a first explicit illustration of the interplay between total, quantum, and classical anticoherence, we consider a rank-2 mixed state of a spin-2 system, corresponding to $N = 2j = 4$ qubits,

$$\rho(\lambda) = \lambda\,|\psi_+\rangle\langle\psi_+| + (1-\lambda)\,|\psi_-\rangle\langle\psi_-|\,, \tag{61}$$

where the orthonormal eigenvectors are given by

$$|\psi_\pm\rangle = \tfrac{1}{2}\big(|2,2\rangle \pm i\sqrt{2}\,|2,0\rangle + |2,-2\rangle\big). \tag{62}$$

The vectors in Eq. (62) span a 1-AC subspace: every pure state in their span is 1-AC [28, 60]. Mixing within this subspace therefore preserves maximal first-order quantum anticoherence. For both the purity and fidelity constructions,

$$\mathcal{A}_1^{T}(\rho) = \mathcal{A}_1^{Q}(\rho) = 1, \qquad \mathcal{A}_1^{C}(\rho) = 0, \tag{63}$$

for every $0 \le \lambda \le 1$. This also shows that classical anticoherence is not a spectral measure of mixedness: $\rho(1/2)$ and the antipodal coherent mixture (19) at $j = 2$ have the same spectrum $(1/2, 1/2, 0, 0, 0)$ and the same first-order reduction $\mathbb{1}_2/2$, yet their classical contributions are zero and one, respectively. The argument applies to all four convex-roof constructions, since every pure state in the support of $\rho(1/2)$ is 1-AC. In particular, a mixed state can have vanishing classical anticoherence.

Higher orders reveal how classical mixing and the choice of measure affect the decomposition. We give the results here; the reduced spectra, convex-roof bounds, and attaining ensembles are derived in Appendix B 1.

Second order: fixed total anticoherence, varying quantum contribution. The two-qubit reduction is $\rho_2 = \mathbb{1}_3/3$ for every λ, so both total measures equal one. Their quantum and classical contributions are

$$\mathcal{A}_2^{Q,\mathcal{P}}(\rho) = 1 - \lambda(1-\lambda),$$
$$\mathcal{A}_2^{C,\mathcal{P}}(\rho) = \lambda(1-\lambda), \tag{64}$$
$$\mathcal{A}_2^{Q,F}(\rho) = \tfrac{1}{2} + \left|\lambda - \tfrac{1}{2}\right|,$$
$$\mathcal{A}_2^{C,F}(\rho) = \tfrac{1}{2} - \left|\lambda - \tfrac{1}{2}\right|. \tag{65}$$

At the pure endpoints, all anticoherence is quantum. As the weights become equal, the total remains maximal while the quantum contribution decreases to $3/4$ for purity and $1/2$ for fidelity. The classical contribution supplies the difference. Thus identical reduced-state isotropy can coexist with different amounts of entanglement, and the two constructions quantify this change differently.

Third order: fixed quantum contribution, varying total anticoherence. For $t = 3$, the roles are reversed: the quantum contributions are independent of λ, while the classical parts vary with the mixing,

$$\mathcal{A}_3^{Q,\mathcal{P}} = \tfrac{2}{3}, \qquad \mathcal{A}_3^{C,\mathcal{P}} = \tfrac{4}{3}\lambda(1-\lambda), \tag{66}$$
$$\mathcal{A}_3^{Q,F} = \tfrac{1}{3}, \qquad \mathcal{A}_3^{C,F} = \tfrac{4}{3}\sqrt{\lambda(1-\lambda)}. \tag{67}$$

The totals follow from $\mathcal{A}_3^{T,X} = \mathcal{A}_3^{Q,X} + \mathcal{A}_3^{C,X}$ for $X = \mathcal{P}, F$. These constant quantum values follow from the first-order values through the complementary-partition relations, Proposition 5 and Eq. (35). They saturate the respective dimensional ceilings, $2/3$ for purity and $1/3$ for fidelity [Eq. (**??**)]; values below unity here do not indicate a loss of maximal entanglement across the 3|1 split. The increase in total anticoherence is entirely classical. Figure 1 compares both constructions at $t = 2, 3$. This example does not establish a universal ordering of the classical contribution with t.

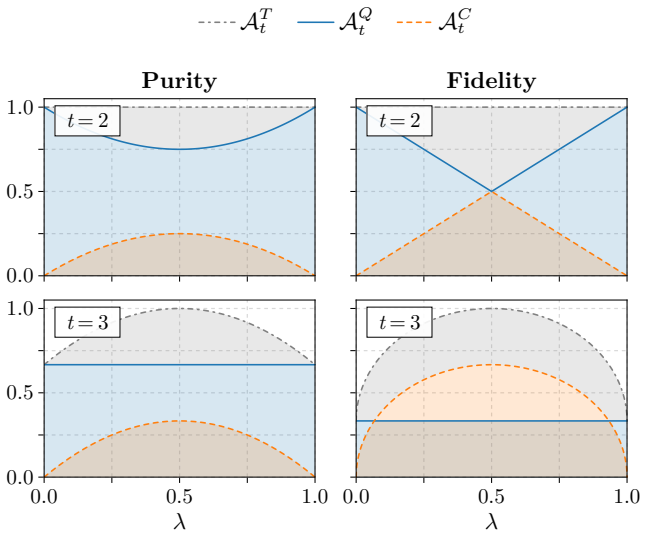

Figure 1. Total (gray dash-dotted), quantum (blue solid), and classical (orange dashed) anticoherence of the spin-2 family in Eq. (61). Columns compare purity and fidelity; rows show $t = 2$ and $t = 3$. All curves are analytic. At $t = 1$, both constructions give $(\mathcal{A}^T, \mathcal{A}^Q, \mathcal{A}^C) = (1, 1, 0)$ for every λ. The quantum values at $t = 3$ saturate their distinct dimensional ceilings. The fidelity total remains a candidate for general covariant channels at higher orders.

A. States saturating the convex-roof negativity

For the quantum AC measure associated with the fidelity-based construction, one must calculate the convex-roof extension of the negativity $\mathcal{N}_t^{\mathrm{CR}}(\rho)$. Here, we discuss two cases where this quantity can be computed explicitly.

1. *Mixed states with support of equal negativity*

If every normalized pure vector in the support of ρ has the same negativity, every decomposition has that average value and the convex roof requires no optimization. In particular, a t-AC subspace consists entirely of pure states with maximal negativity $t/2$ w.r.t. bipartition t|N-t ($t \leq N/2$) [28, 40, 60]. This gives the following characterization of maximal quantum anticoherence.

Theorem 1 (Anticoherent support)**.** *Let ρ have support $S \subset \mathcal{H}_j$ that is a t-AC subspace. Then*

$$\mathcal{N}_t(\rho) = \mathcal{N}_t^{\mathrm{CR}}(\rho) = t/2, \qquad \mathcal{A}_t^{Q,F}(\rho) = \mathcal{A}_t^{Q,\mathcal{P}}(\rho) = 1. \tag{68}$$

Conversely, for either of these quantum constructions, attaining unity implies that the support is a t-AC subspace.

The proof is given in Appendix B 3, using the ordinary-negativity result of Ref. [28]. Such states have total anticoherence one and classical contribution zero: mixing cannot reduce the quantum contribution because every pure state available within the support is already maximally t-AC.

Example 2. 2-AC subspaces for $N = 7$

As a second example, consider the symmetric $N = 7$ qubit states [61, 62]

$$|\psi_1\rangle = \sqrt{\tfrac{3}{10}}\,|\tfrac{7}{2}, \tfrac{7}{2}\rangle + \sqrt{\tfrac{7}{10}}\,|\tfrac{7}{2}, -\tfrac{3}{2}\rangle,$$
$$|\psi_2\rangle = \sqrt{\tfrac{7}{10}}\,|\tfrac{7}{2}, \tfrac{3}{2}\rangle - \sqrt{\tfrac{3}{10}}\,|\tfrac{7}{2}, -\tfrac{7}{2}\rangle. \tag{69}$$

These vectors span a 2-AC subspace. Therefore, for any mixture ρ supported on this subspace one finds

$$\mathcal{N}_1^{\mathrm{CR}}(\rho) = \tfrac{1}{2}, \qquad \mathcal{N}_2^{\mathrm{CR}}(\rho) = 1, \tag{70}$$

and hence $\mathcal{A}_t^{T,F}(\rho) = \mathcal{A}_t^{Q,F}(\rho) = 1$ for $t = 1, 2$.

2. *Orthogonal supports after partial transposition*

A second sufficient condition applies to a decomposition $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$. If the partially transposed projectors $(|\psi_i\rangle\langle\psi_i|)^{T_t}$ have pairwise orthogonal supports, then

$$\mathcal{N}_t^{\mathrm{CR}}(\rho) = \mathcal{N}_t(\rho) = \sum_i p_i \mathcal{N}_t(|\psi_i\rangle). \tag{71}$$

The given decomposition is therefore optimal. The reason is that partial transposition leaves the components in separate orthogonal blocks, so their negative eigenvalues cannot cancel; Appendix B 3 gives the proof. This condition holds for orthogonal bases of AC subspaces [28], but also for supports that are not AC subspaces. An example is provided by the spin-5/2 states

$$\begin{aligned} |\psi_1\rangle &= \tfrac{1}{\sqrt{6}}\,|\tfrac{5}{2}, -\tfrac{5}{2}\rangle + \sqrt{\tfrac{5}{6}}\,|\tfrac{5}{2}, \tfrac{3}{2}\rangle\,, \\ |\psi_2\rangle &= -\sqrt{\tfrac{5}{6}}\,|\tfrac{5}{2}, -\tfrac{3}{2}\rangle + \tfrac{1}{\sqrt{6}}\,|\tfrac{5}{2}, \tfrac{5}{2}\rangle\,. \end{aligned} \tag{72}$$

Across the 1|4 split, the partially transposed projectors have orthogonal supports and each pure state has negativity $\sqrt{2}/3$. Consequently, for $0 \le \lambda \le 1$,

$$\mathcal{N}_1^{\mathrm{CR}}\big(\lambda\,|\psi_1\rangle\langle\psi_1| + (1-\lambda)\,|\psi_2\rangle\langle\psi_2|\big) = \tfrac{\sqrt{2}}{3}. \tag{73}$$

Thus the convex roof can remain constant under mixing even below its maximal value.

V. APPLICATIONS

A. Robustness of quantum AC under particle loss

The four quantum anticoherence measures allow comparison across different particle numbers. Write $\rho_q = \mathrm{Tr}_{N-q}\,\rho_N$, with $t < q \le N$. The identity $(\rho_q)_t = (\rho_N)_t$ leaves every total anticoherence measures unchanged. Each quantum convex roof uses the same concave spectral functional of this t-qubit marginal, with a normalization depending only on t. Tracing out particles from the complementary group is a local operation, so entanglement monotonicity [63] gives

$$\begin{aligned} \mathcal{A}_t^{T,X}(\rho_q) &= \mathcal{A}_t^{T,X}(\rho_N), \\ \mathcal{A}_t^{Q,X}(\rho_q) &\le \mathcal{A}_t^{Q,X}(\rho_N), \\ \mathcal{A}_t^{C,X}(\rho_q) &\ge \mathcal{A}_t^{C,X}(\rho_N). \end{aligned} \tag{74}$$

Thus the loss of quantum contribution is exactly compensated by an increase in the classical contribution. These statements do not require axiom (T4). At $q = t$ the AC property still survives, but the quantum measure lies outside the proper-bipartition domain.

Figure 2(a) compares purity-based quantum anticoherence after particle loss from highest-order anticoherent pure (HOAP) states [40], GHZ states, and W states, for $N = 4, \ldots, 10$. We retain q particles and evaluate the measure only for $t < q \le N$. Partial tracing a symmetric state is also the appropriate description when an unspecified subset of particles is retained; its relation to correlation functions is discussed in Ref. [64].

For GHZ states, every proper reduction is a mixture of two coherent product states, so the quantum contribution vanishes after even one loss. Without loss, the $t = 1, 2$ purity values are 1 and 3/4, respectively, for the sizes considered. For W states the exact expression is

$$\mathcal{A}_t^{Q,\mathcal{P}}(\rho_q) = \frac{2(t+1)(q-t)}{N^2}, \qquad t < q \le N. \tag{75}$$

Indeed, $\rho_q = (q/N)|W_q\rangle\langle W_q| + (1 - q/N)|0\rangle\langle 0|^{\otimes q}$. A pure vector $\alpha|0\rangle^{\otimes q} + \beta|W_q\rangle$ in this support has purity functional $2(t+1)(q-t)|\beta|^4/q^2$. Jensen's inequality gives Eq. (75), attained by two states with $|\beta|^2 = q/N$ and opposite relative phases.

For HOAP state reductions of rank greater than two, the figure reports feasible-ensemble upper bounds on the convex roof. The states, optimization, and residual checks are specified in Appendix C. Pure and rank-two reductions are evaluated exactly. A small upper bound certifies a small quantum contribution to numerical accuracy; a large upper bound alone does not prove robustness. The plot seems to indicate a universal threshold at one-half particle loss for HOAP states. The latter retain a strictly positive quantum contribution for every $q > t$, as certified by Eq. (75).

Figure 2(b) gives the corresponding fidelity-based comparison for the same input states. The GHZ contribution again vanishes after any loss and equals $1/t$ without loss. For W states, the pure-state fidelity functional on the vacuum–W_q support is $2\sqrt{t(q-t)}|\beta|^2/(tq)$. Every decomposition has average $|\beta|^2 = q/N$, giving the exact value $\mathcal{A}_t^{Q,F}(\rho_q) = 2\sqrt{t(q-t)}/(tN)$. Thus both constructions detect the persistence of quantum anticoherence for W states, although their numerical values and dependence on particle number differ. Numerical HOAP fidelity roofs remain feasible-ensemble upper bounds; their evaluation is described in Appendix C.

B. Maximal anticoherence order for mixed states at fixed purity

For mixed states, anticoherence competes with purity: increasing isotropy generally requires mixing, while higher purity constrains the degree to which low-order spin moments can be suppressed. A natural question is therefore the following: *given a fixed global purity, what is the maximal order t of anticoherence that a mixed spin state can achieve?*

This question is particularly relevant from the perspective of quantum resources. At fixed purity, differ-

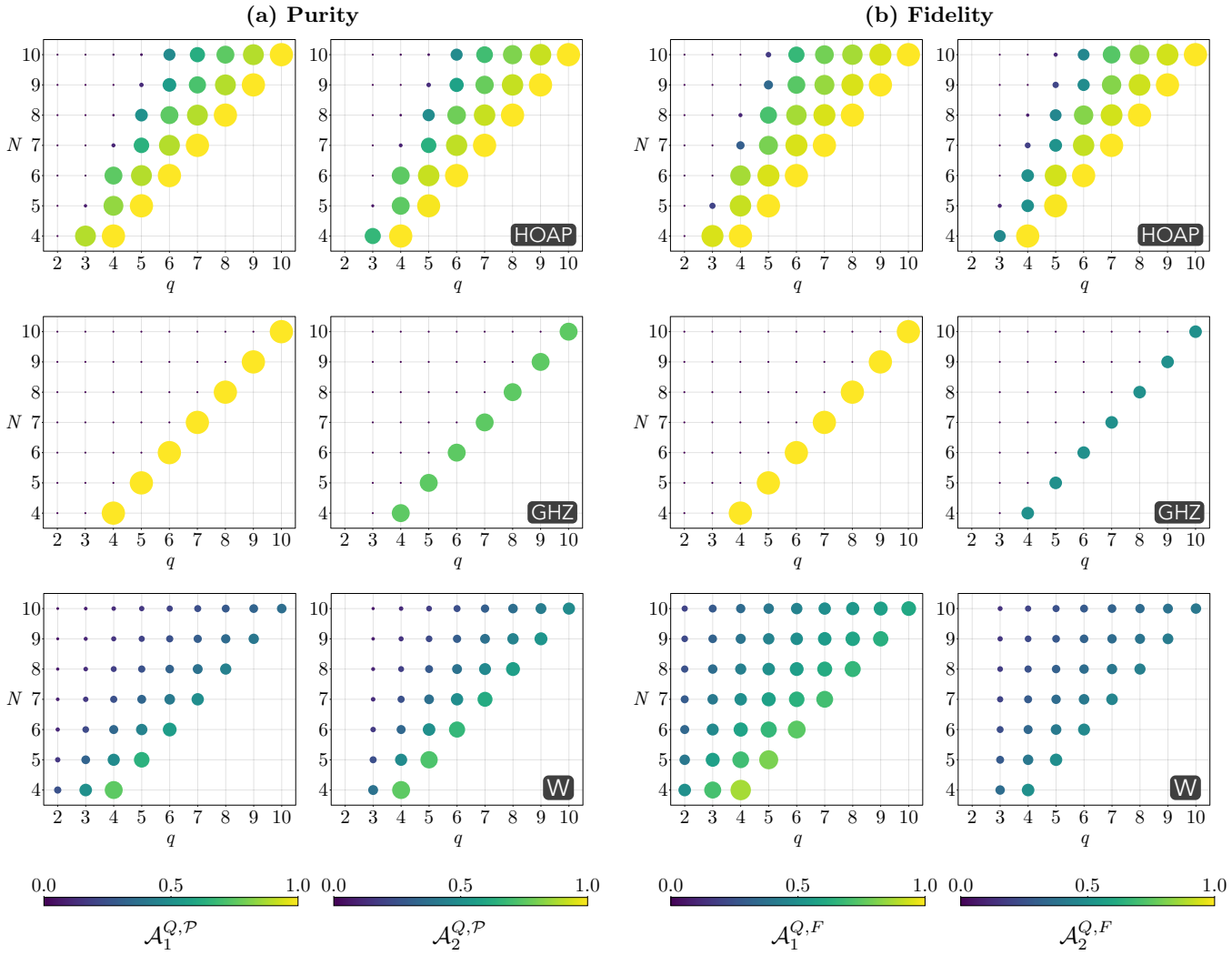

Figure 2. Quantum anticoherence after particle loss: **(a)** purity-based and **(b)** fidelity-based measures for the same input states. Within each block, rows show HOAP states (top), GHZ states (middle), and W states (bottom); columns show $t = 1, 2$. The axes give the initial particle number N and retained number q; only $t < q \leq N$ is shown. Colors encode the displayed values, and disk diameters increase linearly with them from a small minimum size at zero. GHZ and W values are exact in both blocks. For HOAP states, purity values are exact for pure and rank-two reductions; fidelity values are analytic for pure inputs, maximally mixed reductions, and the two-qubit case. All remaining HOAP disks show numerical feasible-ensemble upper bounds on the respective convex roof. The fidelity raw data also give ordinary-negativity lower bounds and the resulting gaps. See Appendix C for the states and numerical methods.

ent states may display the same total amount of isotropy while differing markedly in the relative contributions arising from genuinely quantum correlations and from statistical mixing. Our framework allows one to resolve this distinction by separating total, quantum, and classical contributions to t-anticoherence.

For non-symmetric multipartite systems, related purity–isotropy questions have been studied in Ref. [65]. In the symmetric sector define the convex feasible set and the purity optimum by

$$\mathcal{S}_{N,t} = \left\{\rho \in \mathcal{D}(\mathcal{H}_j) : \rho_t = \rho_0^{(t)}\right\}, \qquad P_{\max}(N,t) = \max_{\rho\in\mathcal{S}_{N,t}} \mathrm{Tr}(\rho^2). \tag{76}$$

The constraints are semidefinite and linear, but maximizing purity is not a semidefinite program: the objective is convex and the maximization is nonconvex. A feasible state proves an attainable purity, not a global upper bound.

If ρ_* is t-AC with purity P_*, the family

$$\rho_s = (1-s)\frac{\mathbb{1}_{N+1}}{N+1} + s\rho_*, \qquad 0 \leq s \leq 1, \tag{77}$$

is t-AC and has purity $1/(N+1) + s^2[P_* - 1/(N+1)]$. Thus an explicit state at a threshold certifies the entire interval below it. Figure 3 shows certified maximal-order curves for $j = 3/2, 2, 5/2$. The point $(1/(N+1), N)$ is the isolated MMS endpoint of the AC-property diagram;

no order-N quantum measure is assigned there.

The explicit examples below establish

$$\begin{aligned} P_{\max}(3,2) &= \tfrac{1}{2}, & P_{\max}(4,3) &= \tfrac{1}{2}, \\ P_{\max}(5,2) &= \tfrac{13}{18}, & P_{\max}(5,3) &= \tfrac{1}{2}, \\ P_{\max}(5,4) &= \tfrac{7}{24}. \end{aligned} \tag{78}$$

The spin-3/2 and spin-2 equalities are proved below. The complete positivity certificates for all three spin-5/2 equalities, together with exact verification programs and an explanation of the methods and their relation to the literature, are provided in the companion repository [66]. Table I gives exact purity roofs for the rank-two examples. For the two independent rank-four roofs, the same repository gives a closed radical formula for q_1 and an exact algebraic characterization of q_2 by a uniquely isolated polynomial root. These quantum/classical divisions are properties of the stated representative states; neither purity nor AC order alone fixes them.

The $\mathcal{P}/F$ stacks in Fig. 3 compare the two constructions for the same representative states. The certified purity thresholds are unchanged because exact t-anticoherence is the same property for both constructions. The quantum and classical allocations, however, depend on the chosen measure. In particular, the fidelity pies must use independently evaluated convex roofs; the exact purity roofs in Table I cannot simply be rescaled into fidelity roofs.

1. $j = \frac{3}{2}$

The first nontrivial example occurs for $j = 3/2$. For pure spin-3/2 states, the maximal anticoherence order is only 1; reaching second-order isotropy therefore requires mixing. Consider the two orthonormal states

$$\begin{aligned} |\psi_1\rangle &= \tfrac{1}{\sqrt{2}}\big(|\tfrac{3}{2},\tfrac{3}{2}\rangle + |\tfrac{3}{2},-\tfrac{1}{2}\rangle\big), \\ |\psi_2\rangle &= \tfrac{1}{\sqrt{2}}\big(-|\tfrac{3}{2},\tfrac{1}{2}\rangle + |\tfrac{3}{2},-\tfrac{3}{2}\rangle\big). \end{aligned} \tag{79}$$

Their equal mixture,

$$\rho = \tfrac{1}{2}\big(|\psi_1\rangle\langle\psi_1| + |\psi_2\rangle\langle\psi_2|\big), \tag{80}$$

has purity 1/2 and is exactly 2-AC. It thus provides the lowest-spin example in which mixing enables a higher anticoherence order than is possible for pure states.

This threshold is globally optimal. A 2-AC spin-3/2 state contains only scalar and rank-three multipoles. With spin time reversal Θ, it obeys $\rho + \Theta\rho\Theta^{-1} = \mathbb{1}_4/2$. Positivity of the second term gives $\rho \le \mathbb{1}_4/2$, hence $\mathrm{Tr}\,\rho^2 \le 1/2$. For purities above 1/2, the maximum order is one; a pure 1-AC state and Eq. (77) make this order attainable throughout the interval.

2. $j = 2$

For $j = 2$, a rank-2 construction already reaches one order higher. Define

$$|\psi_\pm\rangle = \tfrac{1}{2}\big(|2,2\rangle \pm i\sqrt{2}\,|2,0\rangle + |2,-2\rangle\big). \tag{81}$$

The equal mixture

$$\rho = \tfrac{1}{2}\big(|\psi_+\rangle\langle\psi_+| + |\psi_-\rangle\langle\psi_-|\big) \tag{82}$$

has purity 1/2 and is exactly 3-AC. This example is particularly useful because the same two-dimensional support is also a 1-AC subspace, so first-order isotropy remains entirely quantum, while higher-order isotropy contains an increasing classical contribution (see the purity contributions ($\mathcal{P}$) in the middle panel of Fig. 3).

The third-order threshold is also global. If ρ is 3-AC and $\lambda|\psi\rangle\langle\psi|$ is any term in its spectral decomposition, then $\mathbb{1}_4/4 = \rho_3 \ge \lambda\psi_3$. Since ψ_3 has rank at most two, its largest eigenvalue is at least 1/2, and therefore $\lambda \le 1/2$. This holds for every eigenvalue of ρ, giving $\mathrm{Tr}\,\rho^2 \le 1/2$. Pure 2-AC states provide the order-two interval above this threshold. Full order four is possible only at the MMS point.

3. $j = \frac{5}{2}$

For $j = 5/2$, the states below attain the global purity maxima for AC orders two, three, and four (see Fig. 3, right panel).

Rank-2 states. Let

$$\begin{aligned} |\psi_1\rangle &= \tfrac{1}{\sqrt{2}}\big(|\tfrac{5}{2},\tfrac{5}{2}\rangle + |\tfrac{5}{2},-\tfrac{5}{2}\rangle\big), \\ |\psi_2\rangle &= \tfrac{1}{\sqrt{2}}\big(|\tfrac{5}{2},\tfrac{3}{2}\rangle + |\tfrac{5}{2},-\tfrac{3}{2}\rangle\big). \end{aligned} \tag{83}$$

The unequal mixture

$$\rho = \tfrac{1}{6}\,|\psi_1\rangle\langle\psi_1| + \tfrac{5}{6}\,|\psi_2\rangle\langle\psi_2| \tag{84}$$

has purity 13/18 and is exactly 2-AC. It reaches the global second-order purity optimum: the certificate in Ref. [66] proves $\mathrm{Tr}\,\rho^2 \le 13/18$ for every 2-AC spin-5/2 state.

A lower-purity rank-2 mixture reaches third-order anticoherence. Let $|\psi_1\rangle$ and $|\psi_2\rangle$ be the states of Eq. (72).

Then

$$\rho = \tfrac{1}{2}\big(|\psi_1\rangle\langle\psi_1| + |\psi_2\rangle\langle\psi_2|\big) \tag{85}$$

has purity 1/2 and is exactly 3-AC. It reaches the global third-order purity optimum: the certificate in Ref. [66] proves $\mathrm{Tr}\,\rho^2 \le 1/2$ for every 3-AC spin-5/2 state. Compared with the previous state, the additional order of isotropy is obtained by increasing the degree of mixing.

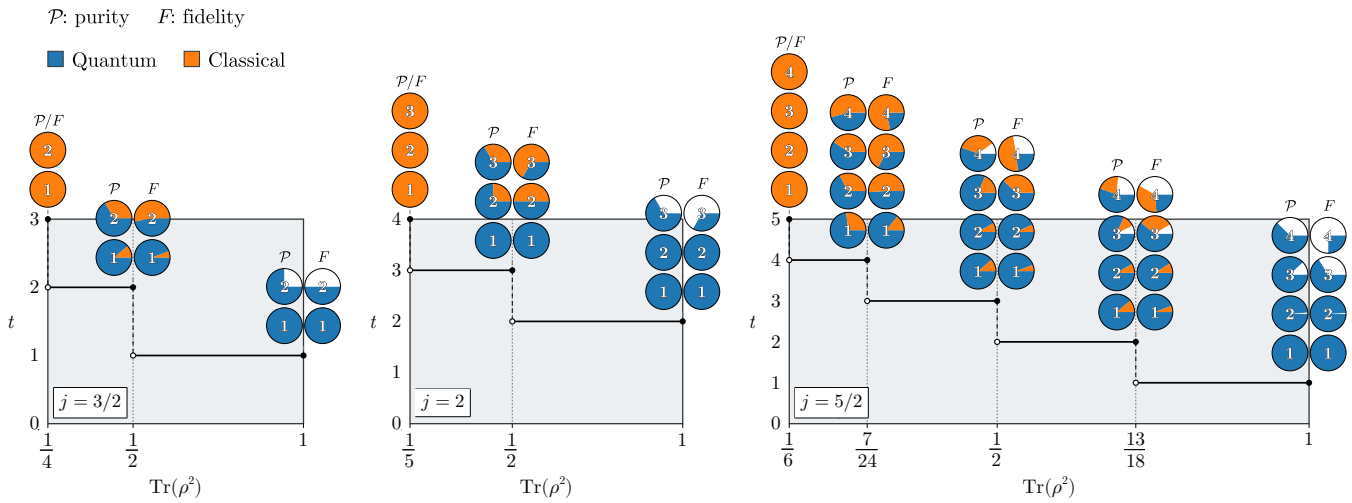

Figure 3. Maximal total AC order versus global purity for $j = 3/2, 2, 5/2$ (left to right). The staircase in each panel applies to both constructions. At the MMS endpoint, a single stack labeled $\mathcal{P}/F$ represents both measures. At the other representative purities, paired pie stacks compare the purity-based measure ($\mathcal{P}$, left) with the fidelity-based measure (F, right). All purity thresholds are globally certified, including 13/18, 1/2, and 7/24 for spin 5/2. Filled dots include the upper-purity endpoints of horizontal segments; open circles exclude the lower-order value at a jump. Dashed vertical lines guide the eye. The isolated MMS points attain order $N = 2j$; no order-N quantum measure is assigned. Stacked pies show orders $t = 1, \ldots, N-1$, labeled at their centers: quantum (blue), classical (orange), and one minus total anticoherence (white). In each panel, the leftmost stack describes the MMS, the rightmost pair HOAP states, and intermediate pairs the states of Table I. All purity-based quantum values are exact, including the algebraic spin-5/2 roofs at purity 7/24; full proofs and certificates are in Ref. [66]. For fidelity, the spin-5/2 blue sectors at purity 7/24, and at purity 13/18 for $t = 2, 3$, are numerical upper bounds on the quantum contribution; orange sectors are the corresponding lower bounds on the classical contribution. All other fidelity quantum values, and all total and white fractions, are analytic. The raw data give both bounds for every numerical entry; see Appendix C.

Rank-4 state. The following rank-four state attains fourth-order anticoherence. Let

$$\begin{aligned} |\psi_1\rangle &= |\tfrac{5}{2}, -\tfrac{3}{2}\rangle, \\ |\psi_2\rangle &= |\tfrac{5}{2}, \tfrac{3}{2}\rangle, \\ |\psi_3\rangle &= \sqrt{\tfrac{9}{20}}\,|\tfrac{5}{2}, \tfrac{5}{2}\rangle + \sqrt{\tfrac{11}{20}}\,|\tfrac{5}{2}, -\tfrac{5}{2}\rangle\,, \\ |\psi_4\rangle &= |\tfrac{5}{2}, -\tfrac{1}{2}\rangle. \end{aligned} \tag{86}$$

The rank-4 mixture

$$\rho = \tfrac{1}{12}\,|\psi_1\rangle\langle\psi_1| + \tfrac{1}{4}\,|\psi_2\rangle\langle\psi_2| + \tfrac{1}{3}\big(|\psi_3\rangle\langle\psi_3| + |\psi_4\rangle\langle\psi_4|\big) \tag{87}$$

has purity 7/24 and is exactly 4-AC. In the basis ordered by $m = 5/2, 3/2, \ldots, -5/2$, it is

$$\rho = \frac{1}{60}\begin{pmatrix} 9 & 0 & 0 & 0 & 0 & \sqrt{99} \\ 0 & 15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 20 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 \\ \sqrt{99} & 0 & 0 & 0 & 0 & 11 \end{pmatrix}. \tag{88}$$

Its nonzero eigenvalues are $1/3, 1/3, 1/4, 1/12$, and direct reduction gives $\rho_4 = \mathbb{1}_5/5$. Any 4-AC spin-5/2 state contains only scalar and rank-five multipoles, so time reversal gives $\rho + \Theta\rho\Theta^{-1} = \mathbb{1}_6/3$. Hence every eigenvalue is at most 1/3, proving $P_{\max}(5,4) \le 1/3$. The explicit sum-of-squares identity in Ref. [66] strengthens this to the exact global bound $P \le 7/24$. Thus this rank-four state attains $P_{\max}(5,4) = 7/24$.

Table I compares the purity- and fidelity-based AC measures order by order for the representative states discussed in Sec. V B.

VI. CONCLUSIONS

We have developed a systematic framework to quantify spin anticoherence beyond the pure-state setting. In its original formulation, spin t-anticoherence is a property of maximal isotropy: a spin state is t-anticoherent when all spin moments up to order t are independent of direction, or equivalently when all multipole moments up to rank t vanish. For mixed states, we recast this condition using the symmetric $N = 2j$ qubit embedding, where it is equivalently expressed by maximal mixedness of the reduced symmetric t-qubit state ρ_t. This reformulation led to an axiomatic separation between *total* t-anticoherence, which quantifies isotropy independently of its origin, and *quantum* t-anticoherence, which isolates the genuinely quantum contribution as a resource monotone associated with a chosen total measure. Their difference defines a *classical* contribution, capturing isotropy

j	State	Rank	$\mathrm{Tr}(\rho^2)$	t	Purity-based measure[a] $(\mathcal{A}_t^{T,\mathcal{P}}, \mathcal{A}_t^{Q,\mathcal{P}}, \mathcal{A}_t^{C,\mathcal{P}})$	Fidelity-based measure[b,c] $(\mathcal{A}_t^{T,F}, \mathcal{A}_t^{Q,F}, \mathcal{A}_t^{C,F})$
$\frac{3}{2}$	(80)	2	$\frac{1}{2}$	1	$(1, \frac{8}{9}, \frac{1}{9})$	$(1, \alpha, 1-\alpha)$
				2	$(1, \frac{2}{3}, \frac{1}{3})$	$(1, \frac{\alpha}{2}, 1-\frac{\alpha}{2})$
2	(82)	2	$\frac{1}{2}$	1	$(1, 1, 0)$	$(1, 1, 0)$
				2	$(1, \frac{3}{4}, \frac{1}{4})$	$(1, \frac{1}{2}, \frac{1}{2})$
				3	$(1, \frac{2}{3}, \frac{1}{3})$	$(1, \frac{1}{3}, \frac{2}{3})$
$\frac{5}{2}$	(84)	2	$\frac{13}{18}$	1	$(1, \frac{8}{9}, \frac{1}{9})$	$(1, \alpha, 1-\alpha)$
				2	$(1, \frac{11}{12}, \frac{1}{12})$	$(1, r, 1-r)$
				3	$(\frac{25}{27}, \frac{22}{27}, \frac{1}{9})$	$(\tau, \frac{2}{3}r, \tau-\frac{2}{3}r)$
				4	$(\frac{55}{72}, \frac{5}{9}, \frac{5}{24})$	$(\frac{7}{12}, \frac{\alpha}{4}, \frac{7}{12}-\frac{\alpha}{4})$
$\frac{5}{2}$	(85)	2	$\frac{1}{2}$	1	$(1, \frac{8}{9}, \frac{1}{9})$	$(1, \alpha, 1-\alpha)$
				2	$(1, \frac{11}{12}, \frac{1}{12})$	$(1, \beta, 1-\beta)$
				3	$(1, \frac{22}{27}, \frac{5}{27})$	$(1, \frac{2}{3}\beta, 1-\frac{2}{3}\beta)$
				4	$(\frac{65}{72}, \frac{5}{9}, \frac{25}{72})$	$(\frac{1}{4}+\frac{\alpha}{2}, \frac{\alpha}{4}, \frac{1}{4}+\frac{\alpha}{4})$
$\frac{5}{2}$	(87)	4	$\frac{7}{24}$	1	$(1, q_1, 1-q_1)$	$(1, f_1, 1-f_1)$
				2	$(1, q_2, 1-q_2)$	$(1, f_2, 1-f_2)$
				3	$(1, \frac{8}{9}q_2, 1-\frac{8}{9}q_2)$	$(1, \frac{2}{3}f_2, 1-\frac{2}{3}f_2)$
				4	$(1, \frac{5}{8}q_1, 1-\frac{5}{8}q_1)$	$(1, \frac{1}{4}f_1, 1-\frac{1}{4}f_1)$

[a] For the rank-four state (87), $q_t = \mathcal{A}_t^{Q,\mathcal{P}}(\rho)$, with $q_1 = (694 - 4\sqrt{22} - 2\sqrt{6071 + 448\sqrt{22}})/675 \approx 0.7325$ and $q_2 = 171/400 + (u + 4v)/12 \approx 0.6706$. Here u, v belong to the uniquely isolated solution of the four-variable polynomial system in Ref. [66].

[b] $\alpha = 2\sqrt{2}/3$, $\beta = (\sqrt{6} + \sqrt{3} + \sqrt{2})/6$, and $\tau = (3 + \sqrt{5} + 2\sqrt{3} + 2\sqrt{15})/18$.

[c] The fidelity roofs r (order two, state (84)) and f_1, f_2 (orders one and two, state (87)) satisfy $(6 + \sqrt{13})/12 \le r \le (1 + \sqrt{3})/3$, $q_1 \le f_1 \lesssim 0.8554$, and $\frac{2}{3}q_2 \le f_2 \lesssim 0.4909$. These three roofs are not determined exactly here. The $\lesssim$ endpoints are numerical feasible-ensemble upper estimates, not interval-certified bounds.

Table I. Total, quantum, and classical AC contributions for the mixed spin states of Sec. V B, compared at each order t. Each block specifies the spin j, the equation defining the state, its rank, and its global purity. For either measure, a unit total contribution is equivalent to t-anticoherence. All purity entries are exact: see Appendix B 2 for the rank-two results and Ref. [66] for the rank-four roofs, their positivity certificates and attaining ensembles. The latter reference also proves that all three spin-5/2 states maximize purity at their highest AC order. All fidelity total entries and all quantum entries independent of r, f_1, f_2 are exact. Classical contributions equal total minus quantum, so their bounds are reversed. Complementary orders follow from Proposition 5 for purity and Eq. (35) for fidelity. See Appendix C for the fidelity evaluations and bounds and Fig. 3 for both graphical decompositions.

produced by statistical mixing.

On the constructive side, we introduced broad families of total measures based on normalized distances from ρ_t to the MMS state, including a Hilbert–Schmidt distance-based measure and a fidelity-based total candidate, as well as simple purity-based and cumulative-multipole measures. The total monotonicity of the fidelity-based candidate under all SU(2)-covariant channels remains supported by numerical evidence. We also showed how cumulative multipole functionals fit naturally into the same axiomatic picture, thereby clarifying the relation with earlier tensor-based descriptions of angular structure. For the quantum contribution, we constructed convex-roof extensions from pure-state functionals tied to bipartite entanglement across the $t \,|\, N-t$ split, obtaining explicit purity/linear-entropy-based quantum measures and a fidelity/convex-roof-negativity construction satisfying the quantum-side requirements and the ordering relative to the fidelity-based total candidate. For the constructions where the required ordering is established, the inequality $\mathcal{A}_t^Q \le \mathcal{A}_t^T$ ensures a well-defined and non-negative classical part $\mathcal{A}_t^C = \mathcal{A}_t^T - \mathcal{A}_t^Q$.

Our examples illustrate that mixed-state anticoherence exhibits qualitatively new behavior absent in the pure-state regime. In particular, mixed states supported on t-AC subspaces can retain *maximal quantum* t-anticoherence independently of the mixing weights, showing that perfect isotropy need not be of classical origin. Conversely, when the achievable anticoherence order is increased at fixed purity, the total isotropy typically becomes more classical: higher-order isotropy is accessible only at sufficiently low purity, and its quantum share generally decreases. We also showed contrasting robustness patterns under particle loss, with HOAP, GHZ, and W states displaying markedly different decay profiles of the quantum contribution.

A conceptual message of this work is that *isotropy of quantum origin* can itself be viewed as a resource, com-

plementary to the usual resource-theoretic viewpoint in which SU(2)-*asymmetry* is the resource. In the asymmetry framework, useful states are those that break rotational symmetry and carry directional information. Here, by contrast, the quantum measure $\mathcal{A}_t^Q$ isolates the part of isotropy that cannot be reproduced by classical randomization over directions. It therefore identifies states that are direction-insensitive for intrinsically quantum reasons, encoded in genuine multipartite correlations in the symmetric sector. This entanglement-based perspective is already built into our axioms, through the requirements of vanishing on separable symmetric states and monotonicity under LOCC, and is made concrete by the explicit measures introduced here. The resulting distinction is operationally relevant whenever one wishes to separate loss of directional information caused by uncontrolled classical noise from isotropy engineered coherently through entanglement.

Several directions follow naturally. It would be interesting to sharpen the operational role of total, quantum, and classical anticoherence in concrete reference-frame alignment and direction-estimation protocols, and to characterize optimal states under constraints such as purity, rank, or particle loss. On the structural side, a broader classification of t-AC subspaces and of mixed states saturating the convex-roof constructions would clarify the landscape of extremal isotropic resources. Finally, although our analysis focuses on permutation-symmetric sectors and SU(2), the same strategy should extend to other symmetry groups and multipartite settings, providing a general route to separate intrinsically quantum isotropy from isotropy generated by classical uncertainty.

We have formulated a distinction between total reduced-state isotropy and an entanglement-based quantum contribution to mixed-state anticoherence, on the proper-reduction domain $1 \le t < N$. For concave totals and their convex roofs, the classical contribution is the maximal increase in isotropy due to mixing pure states. Equivalently, it is the maximal average reduction in marginal isotropy obtained by measuring a purifying environment and learning the outcome. This is a property of the state and the chosen functional, independent of the actual preparation.

The purity and Hilbert–Schmidt constructions satisfy the full total and quantum axioms. Trace distance and fidelity offer discrimination- and overlap-based motivations; their total monotonicity is proved at first order and for random rotations, while the general higher-order covariant-channel statement remains open. Cumulative multipoles provide another total description, but the spin-2 counterexample prevents a general LOCC-monotone extension with the same pure-state values. This obstruction has exceptions, including the spin-3/2, order-two case discussed above.

Anticoherent subspaces connect maximal quantum anticoherence to existing results on mixed rotosensors. The first-order purity roof gives a bound on the sum of rotational QFIs, with an exact characterization at saturation. Our explicit spin-2 example also shows that different higher-order quantum measures can vary while the rotational QFI is unchanged. No single measure introduced here is therefore a universal metrological performance score.

Under particle loss, the GHZ and W formulas are exact, whereas higher-rank HOAP results are numerical upper bounds. For the purity–order trade-off, we prove the relevant thresholds for spins 3/2 and 2, and certify the global optima of orders two, three, and four for spin 5/2. For the rank-four spin-5/2 example, a closed radical formula gives the first-order purity roof, and a uniquely isolated polynomial root specifies the second-order roof exactly. Matching global positivity certificates and attaining ensembles prove both results. The higher-rank HOAP roofs remain open. These questions, and the general covariant-noise monotonicity of the trace-distance and fidelity totals, delimit the claims supported by the present framework.

ACKNOWLEDGEMENTS

JM and ESE thank Nicolas Cerf for stimulating discussions on this topic. This work is supported by the Quantum Science and Technology - National Science and Technology Major Project (Grant No. 2025ZD0300801) and the National Natural Science Foundation of China (Grant Nos. 92476201). JM and ESE acknowledge the FWO and the F.R.S.-FNRS for their funding as part of the Excellence of Science programme (EOS project 40007526).

Appendix A: Proofs

We begin this Appendix on the proofs of our propositions by two useful results. A general observation is that the ordering axiom (Q6) follows automatically whenever the chosen total measure is concave.

Proposition 10. *Let $\mathcal{A}_t^T : \mathcal{D}(\mathcal{H}_j) \to [0,1]$ be a total t-anticoherence measure whose restriction to pure states is used to define*

$$\mathcal{A}_t^Q(\rho) = \min_{\{p_i, |\psi_i\rangle\}} \sum_i p_i \, \mathcal{A}_t^T(|\psi_i\rangle\langle\psi_i|). \tag{A1}$$

If $\mathcal{A}_t^T$ is concave on $\mathcal{D}(\mathcal{H}_j)$, then

$$\mathcal{A}_t^Q(\rho) \le \mathcal{A}_t^T(\rho) \qquad \forall\, \rho \in \mathcal{D}(\mathcal{H}_j), \tag{A2}$$

i.e., axiom (Q6) holds.

Proof. Let $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ be any pure-state decomposition. By concavity of $\mathcal{A}_t^T$,

$$\mathcal{A}_t^T(\rho) \ge \sum_i p_i \, \mathcal{A}_t^T(|\psi_i\rangle\langle\psi_i|). \tag{A3}$$

Since this inequality holds for every decomposition, it also holds for the one minimizing the convex roof, which gives

$$\mathcal{A}_t^T(\rho) \geq \mathcal{A}_t^Q(\rho). \tag{A4}$$

□

We will also need a result derived in [63].

Theorem 2 ([63]). *Let $f : \mathcal{D}(\mathcal{H}_{t/2}) \to \mathbb{R}$ be concave and invariant under unitary operations. Then the pure-state functional*

$$A_t^Q(|\psi\rangle) = f\big(\operatorname{Tr}_{N-t}(|\psi\rangle\langle\psi|)\big) \tag{A5}$$

is non-increasing on average under LOCC operations across the bipartition $t\,|\,N{-}t$. Its convex-roof extension,

$$A_t^Q(\rho) = \min_{\{p_i,|\psi_i\rangle\}} \sum_i p_i A_t^Q(|\psi_i\rangle), \tag{A6}$$

is non-increasing under LOCC operations across that bipartition.

1. Proof of Prop. 1

Proof. First, $\operatorname{Tr}(\rho_t^2) = 1/(t+1)$ if and only if ρ_t is maximally mixed, which gives (T1).

Let us now prove (T2). If ρ is a pure coherent state, then every reduced state ρ_t is pure, so $\operatorname{Tr}(\rho_t^2) = 1$ and hence $\mathcal{A}_t^{T,\mathcal{P}}(\rho) = 0$. Conversely, if $\mathcal{A}_t^{T,\mathcal{P}}(\rho) = 0$, then $\operatorname{Tr}(\rho_t^2) = 1$, so ρ_t is pure. Since ρ_t is obtained by tracing out part of the symmetric N-qubit state ρ_S, the purity of ρ_t implies that ρ_S is product across the bipartition $t\,|\,N{-}t$. Because ρ_S is supported on the fully symmetric subspace, separability across this bipartition is equivalent to full separability [50]; hence ρ_S is a convex mixture of symmetric product states. The fact that ρ_t is pure then forces this mixture to contain only one product component, so $\rho_S = (|\mathbf{n}\rangle\langle\mathbf{n}|)^{\otimes N}$, i.e. ρ is a pure spin-coherent state. This proves (T2).

The SU(2) invariance (T3) is inherited from the invariance of $\operatorname{Tr}(\rho_t^2)$ under global rotations, and normalization to $[0,1]$ is ensured by the prefactor $(t+1)/t$.

We now prove monotonicity under SU(2)-covariant noise (T4). Let Φ be an SU(2)-covariant channel on $\mathcal{H}_j$. By covariance, the multipole coefficients transform as $\rho_{LM} \mapsto f_L \rho_{LM}$ with $|f_L| \leq 1$. Using the Hilbert–Schmidt normalization of the tensors $T^{(t)}_{LM}$, we obtain

$$\begin{aligned}\operatorname{Tr}([(\Phi(\rho))_t]^2) &= \frac{1}{t+1} + \sum_{L=1}^{t} c_{N,t,L}^2 f_L^2 \sum_{M=-L}^{L} |\rho_{LM}|^2 \\ &\leq \frac{1}{t+1} + \sum_{L=1}^{t} c_{N,t,L}^2 \sum_{M=-L}^{L} |\rho_{LM}|^2 \\ &= \operatorname{Tr}(\rho_t^2).\end{aligned} \tag{A7}$$

Hence $\mathcal{A}_t^{T,\mathcal{P}}(\Phi(\rho)) \geq \mathcal{A}_t^{T,\mathcal{P}}(\rho)$, proving (T4). □

2. Proof of Prop. 2

Proof. Axioms (T1)–(T3) follow from the distance-based construction. Equation (25) writes the Hilbert–Schmidt total as the increasing function $x \mapsto 1 - \sqrt{1-x}$ of the purity total. Its monotonicity (T4) therefore follows immediately from Proposition 1.

For its quantum extension, the reduced-state functional

$$f(\sigma) = 1 - \sqrt{\frac{t+1}{t}}\,\|\sigma - \mathbb{1}_{t+1}/(t+1)\|_2 \tag{A8}$$

is concave and unitarily invariant. Theorem 2 gives (Q5), and Proposition 10 gives (Q6). The other quantum axioms follow directly as for purity. The same concavity argument applies to the trace-norm quantum construction, independently of the unresolved higher-order total monotonicity. □

3. Proof of Prop. 3

Proof. From the Schmidt decomposition, it can be seen that the reduced state of $|\psi\rangle$ on A is given by

$$\rho_A = \sum_{i=1}^{d_A} \alpha_i^2 \,|\phi_A^{(i)}\rangle\,\langle\phi_A^{(i)}|\,, \tag{A9}$$

So, the eigenvalues λ_i of ρ_A are related to the Schmidt coefficients α_i, that is, $\lambda_i = \alpha_i^2$. For pure states $\rho = |\psi\rangle\langle\psi|$, the fidelity between ρ_A and the MMS verifies

$$F(\rho_A, \rho_0) = \frac{1}{d_A}\left(\operatorname{Tr}\sqrt{\rho_A}\right)^2 = \frac{1}{d_A}\left(\sum_{i=1}^{d_A}\sqrt{\lambda_i}\right)^2. \tag{A10}$$

The eigenvalues of the reduced state ρ_A verify

$$\left(\sum_{i=1}^{d_A}\sqrt{\lambda_i}\right)^2 = \underbrace{\sum_{i=1}^{d_A}\lambda_i}_{=1} + 2\sum_{i>j=1}^{d_A}\sqrt{\lambda_i\lambda_j} \tag{A11}$$

and from (31), we see that [54]

$$\left(\sum_{i=1}^{d_A}\sqrt{\lambda_i}\right)^2 = 1 + 2\mathcal{N}_A(|\psi\rangle). \tag{A12}$$

Injecting (A12) in (A10), we obtain our result (32). □

4. Proof of Prop. 4

Proof. Pure-state consistency (Q2) and SU(2) invariance (Q3) are immediate from the definition and the rotational covariance of the reduced state. Convexity (Q4) follows directly from the convex-roof construction. Axiom (Q5)

follows from Theorem 2 since the function $f(\rho) = 1 - \mathrm{Tr}(\rho^2)$ is concave and invariant under unitaries due to the properties of the purity. Axiom (Q1) follows because fully separable symmetric states admit decompositions into coherent states, and $\mathcal{A}_t^{\mathcal{P}}$ vanishes on pure coherent states.

To prove (Q6), it is enough to note that the total purity-based measure $\mathcal{A}_t^{T,\mathcal{P}}(\rho) = \frac{t+1}{t}\big(1 - \mathrm{Tr}(\rho_t^2)\big)$ is concave (equivalently, $-\mathrm{Tr}(\rho_t^2)$ is concave because $\mathrm{Tr}(\rho_t^2)$ is convex and $\rho \mapsto \rho_t$ is linear). Therefore the ordering follows directly from the general convex-roof ordering Proposition 10. □

5. Proof of Prop. 5

Proof. For pure states $|\psi\rangle$, the reduced density matrices across the bipartitions $t\,|\,N{-}t$ and $(N{-}t)\,|\,t$ have identical nonzero spectra, which implies $\mathrm{Tr}(\rho_t^2) = \mathrm{Tr}(\rho_{N-t}^2)$ and therefore

$$\mathcal{A}_t^{\mathcal{P}}(|\psi\rangle) = c_t\, \mathcal{A}_{N-t}^{\mathcal{P}}(|\psi\rangle), \quad c_t = \frac{(t+1)(N-t)}{t(N+1-t)}. \tag{A13}$$

Let $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ be an arbitrary pure-state decomposition. Using the above relation term by term yields

$$\sum_i p_i\, \mathcal{A}_t^{\mathcal{P}}(|\psi_i\rangle) = c_t \sum_i p_i\, \mathcal{A}_{N-t}^{\mathcal{P}}(|\psi_i\rangle). \tag{A14}$$

Since this equality holds for every decomposition, it also holds for the minimizing decompositions defining the convex-roof extensions, giving

$$\mathcal{A}_t^{Q,\mathcal{P}}(\rho) = c_t\, \mathcal{A}_{N-t}^{Q,\mathcal{P}}(\rho), \tag{A15}$$

which proves the purity relation. For fidelity, the two complementary reductions have the same nonzero spectra, so their pure-state negativities coincide. Multiplication by the normalization factors $2/t$ and $2/(N-t)$, followed by the same ensemble minimization, proves Eq. (35). □

6. Proof of Prop. 6

Proof. Axioms (Q2) and (Q4) follow directly from the convex-roof construction, while axiom (Q3) follows from the SU(2) invariance of the pure-state functional. Axiom (Q5) follows by assumption, since $\mathcal{A}_t^d(|\psi\rangle)$ is an entanglement monotone across the bipartition $t\,|\,N{-}t$ and convex roofs of pure-state entanglement monotones are LOCC monotones. Axiom (Q1) follows because the pure-state functional vanishes on coherent states, and fully separable symmetric states admit decompositions into coherent states. Finally, if $\mathcal{A}_t^{T,d}$ is concave, axiom (Q6) follows from Proposition 10. □

7. Proof of Prop. 7

Proof. We first recall the definition of the total fidelity-based measure:

$$\mathcal{A}_t^{T,F}(\rho) = \frac{(\mathrm{Tr}(\sqrt{\rho_t}))^2 - 1}{t} = \frac{(t+1)F(\rho_t, \rho_0^{(t)}) - 1}{t}. \tag{A16}$$

As discussed in Sec. III A 3, this quantity satisfies axioms (T1)–(T3). Indeed, (T1) follows from $F(\rho_t, \rho_0^{(t)}) = 1$ iff $\rho_t = \rho_0^{(t)}$, while (T2) follows from the fact that $F(\rho_t, \rho_0^{(t)}) = 1/(t+1)$ iff ρ_t is pure, which in the symmetric sector forces the global state to be a pure spin-coherent state. Axiom (T3) follows from unitary invariance of fidelity.

We now prove the quantum axioms for

$$\mathcal{A}_t^{Q,F}(\rho) = \frac{2}{t}\mathcal{N}_t^{\mathrm{CR}}(\rho). \tag{A17}$$

Axiom (Q1) holds because every fully separable symmetric state admits a decomposition into coherent states, and the negativity vanishes on each coherent pure state. Axiom (Q3) follows because global rotations act by local unitaries with respect to the bipartition $t\,|\,N{-}t$, and therefore do not change the negativity. Convexity (Q4) follows directly from the convex-roof construction, and monotonicity under LOCC operations (Q5) follows from the corresponding property of the convex-roof negativity [55].

It remains to verify pure-state consistency (Q2) and the ordering axiom (Q6). For pure states, Proposition 3 gives

$$F(\rho_t, \rho_0^{(t)}) = \frac{1 + 2\mathcal{N}_t(|\psi\rangle)}{t+1}. \tag{A18}$$

Therefore,

$$\begin{aligned} \mathcal{A}_t^{T,F}(|\psi\rangle) &= \frac{(t+1)F(\rho_t, \rho_0^{(t)}) - 1}{t} \\ &= \frac{2}{t}\mathcal{N}_t(|\psi\rangle) = \mathcal{A}_t^{Q,F}(|\psi\rangle), \end{aligned} \tag{A19}$$

which proves (Q2).

The ordering axiom (Q6) follows from concavity. A short variational identity makes this explicit:

$$(\mathrm{Tr}\,\sqrt{X})^2 = \inf_{Y>0,\,\mathrm{Tr}\,Y=1} \mathrm{Tr}(XY^{-1}), \qquad X \geq 0. \tag{A20}$$

The Hilbert–Schmidt Cauchy–Schwarz inequality gives $(\mathrm{Tr}\,\sqrt{X})^2 \leq \mathrm{Tr}(XY^{-1})\,\mathrm{Tr}\,Y$. Equality holds for $Y = \sqrt{X}/\,\mathrm{Tr}\,\sqrt{X}$ when $X > 0$; singular X follow by a limit. As an infimum of linear functions of X, the right-hand side is concave. Thus $\mathcal{A}_t^{T,F}$ is concave, and Proposition 10 yields

$$\mathcal{A}_t^{Q,F}(\rho) \leq \mathcal{A}_t^{T,F}(\rho). \tag{A21}$$

□

8. Proof of Prop. 8

Proof. We first show that $\mathcal{A}_t^{T,\mathrm{cm}}(\rho)$ is indeed a function taking values in $[0,1]$, which is equivalent to proving that $0 \le C_{\le t}(\rho) \le C_{\le t}(\rho_{\mathrm{coh}})$. Since it is a sum of positive quantities, it is immediate to verify that $C_{\le t}(\rho) \ge 0$. The upper bound is proved in the following lemma:

Lemma 2. *For every spin-j density matrix ρ and every $t = 1, \dots, 2j-1$,*

$$C_{\le t}(\rho) \le C_{\le t}(\rho_{\mathrm{coh}}), \tag{A22}$$

where ρ_{coh} is any spin-coherent state.

Proof. For pure states, the cumulative-multipole extremality result gives the bound, with equality only for spin-coherent states up to rotations when $1 \le t < N$ [3].

For mixed states, write $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$. Since $C_{\le t}$ is the squared Hilbert–Schmidt norm of the projection of ρ onto the subspace spanned by the tensor operators T_{LM} with $1 \le L \le t$, it is convex. Hence

$$C_{\le t}(\rho) \le \sum_i p_i C_{\le t}(|\psi_i\rangle\langle\psi_i|) \le C_{\le t}(\rho_{\mathrm{coh}}). \tag{A23}$$

□

For completeness, no additional mixed maximizers arise. Equality in the two inequalities above requires every pure component to be coherent and their projected multipole vectors to coincide, by strict convexity of the squared Euclidean norm. Since this projection contains the rank-one components, all these coherent states have the same mean-spin direction and are the same state. Thus the mixed maximizer is itself pure and coherent.

We now verify the defining properties of $\mathcal{A}_t^{T,\mathrm{cm}}(\rho)$. Axiom (T1) follows directly from the affine normalization in Eq. (50): by definition, $C_{\le t}(\rho) = 0$ if and only if all multipoles up to rank t vanish, i.e. if and only if ρ is t-AC. For axiom (T2), $\mathcal{A}_t^{T,\mathrm{cm}}(\rho) = 0$ if and only if ρ is a pure coherent state, using the coherent-state extremality in Lemma 2 together with the uniqueness of the pure-state maximizers. Axiom (T3) holds because under a global rotation the coefficients ρ_{LM} transform unitarily within each fixed L, so each $a_L(\rho) \equiv \sum_{M=-L}^{L} |\rho_{LM}|^2$ is invariant, hence so is $C_{\le t}(\rho)$.

It remains to verify axiom (T4). Let Φ be an SU(2)-covariant channel on $\mathcal{H}_j$. By the standard structure theorem for covariant channels on a fixed spin irrep (see Sec. II B around Eq. (10)), Φ acts diagonally on irreducible tensor operators:

$$\Phi(T_{LM}) = f_L\, T_{LM}, \tag{A24}$$

with f_L independent of M and satisfying $|f_L| \le 1$. Therefore, if

$$\rho = \frac{\mathbb{1}}{2j+1} + \sum_{L=1}^{2j} \sum_{M=-L}^{L} \rho_{LM} T_{LM}, \tag{A25}$$

then for $\rho' = \Phi(\rho)$ one has $\rho'_{LM} = f_L \rho_{LM}$, and thus

$$\sum_{M=-L}^{L} |\rho'_{LM}|^2 = |f_L|^2 \sum_{M=-L}^{L} |\rho_{LM}|^2 \le \sum_{M=-L}^{L} |\rho_{LM}|^2. \tag{A26}$$

Summing over $L = 1, \dots, t$ gives

$$C_{\le t}(\Phi(\rho)) \le C_{\le t}(\rho). \tag{A27}$$

Since $\mathcal{A}_t^{T,\mathrm{cm}}$ is an affine decreasing function of $C_{\le t}$, it follows that

$$\mathcal{A}_t^{T,\mathrm{cm}}(\Phi(\rho)) \ge \mathcal{A}_t^{T,\mathrm{cm}}(\rho), \tag{A28}$$

which is axiom (T4). □

9. Proof of Prop. 9

Proof. For every spin-j state,

$$C_{\le 1}(\rho) = \frac{3|\langle \mathbf{J} \rangle|^2}{j(j+1)(2j+1)}. \tag{A29}$$

The coherent value is obtained by setting $|\langle \mathbf{J} \rangle| = j$. The normalized cumulative total is therefore $1 - |\langle \mathbf{J} \rangle|^2/j^2$, which equals $\mathcal{A}_1^{T,\mathcal{P}}(\rho)$ for mixed as well as pure states.

Taking convex roofs of the same pure-state functional gives

$$\mathcal{A}_1^{Q,\mathrm{cm}}(\rho) = \mathcal{A}_1^{Q,\mathcal{P}}(\rho). \tag{A30}$$

Since $\mathcal{A}_1^{Q,\mathcal{P}}$ satisfies axioms (Q1)–(Q6) by Proposition 4, the same is true for $\mathcal{A}_1^{Q,\mathrm{cm}}$, in particular (Q5). □

Appendix B: Additional analytic evaluations

1. Analytic evaluation of the spin-2 family

We derive the results of Example 1 in Sec. IV for the family $\rho(\lambda)$ defined in Eq. (61).

a. *First order*

Every pure vector in the support is 1-AC, so both first-order quantum measures equal one. This also follows directly for the fidelity construction from the orthogonal supports of $(|\psi_\pm\rangle\langle\psi_\pm|)^{T_1}$:

$$\mathcal{N}_1(\rho) = \lambda \mathcal{N}_1(|\psi_+\rangle) + (1-\lambda)\mathcal{N}_1(|\psi_-\rangle) = \tfrac{1}{2}. \tag{B1}$$

The ordinary negativity is a lower bound on its convex roof, and this decomposition attains it. Hence $\mathcal{N}_1^{\mathrm{CR}}(\rho) = 1/2$ and $\mathcal{A}_1^{Q,F}(\rho) = 1$.

b. Second-order purity roof

The two-qubit reduction of each eigenvector is $\mathbb{1}_3/3$, and therefore $\rho_2(\lambda) = \mathbb{1}_3/3$. To evaluate the quantum contribution, we optimize over pure-state decompositions of $\rho(\lambda)$. We identify the span of $\{|\psi_+\rangle, |\psi_-\rangle\}$ with an effective qubit and write ρ in Bloch form with $\mathbf{r} = (0, 0, r_z)$, $r_z = 2\lambda - 1$. Any decomposition $\rho = \sum_i p_i |\phi_i\rangle\langle\phi_i|$ corresponds to an ensemble of pure Bloch vectors $\mathbf{n}_i$ with $\sum_i p_i \mathbf{n}_i = \mathbf{r}$, and therefore $\sum_i p_i z_i = r_z$, where z_i denotes the z-component of $\mathbf{n}_i$.

For a pure state $|\phi\rangle$ in the support, the purity-based functional can be written in terms of its Bloch z-component as

$$\mathcal{A}_2^{\mathcal{P}}(|\phi\rangle) = \frac{3 + z^2}{4}. \tag{B2}$$

Hence, for an arbitrary decomposition,

$$\begin{aligned} \sum_i p_i \mathcal{A}_2^{\mathcal{P}}(|\phi_i\rangle) &= \frac{3}{4} + \frac{1}{4}\sum_i p_i z_i^2 \\ &\geq \frac{3}{4} + \frac{1}{4}\Big(\sum_i p_i z_i\Big)^2 = \frac{3 + r_z^2}{4}, \end{aligned} \tag{B3}$$

where we used Jensen's inequality for the convex function z^2. The bound is tight. For $r_z = 2\lambda - 1$, choose θ such that $\cos\theta = r_z$ and define

$$|\phi_\pm\rangle = \cos\frac{\theta}{2}|\psi_+\rangle \pm \sin\frac{\theta}{2}|\psi_-\rangle. \tag{B4}$$

Then $\rho = \frac{1}{2}|\phi_+\rangle\langle\phi_+| + \frac{1}{2}|\phi_-\rangle\langle\phi_-|$, and both $|\phi_\pm\rangle$ have the same Bloch z-component $z_\pm = r_z$, so the Jensen bound is saturated. Therefore,

$$\mathcal{A}_2^{\mathcal{Q},\mathcal{P}}(\rho) = \frac{3 + r_z^2}{4} = 1 - \lambda(1 - \lambda). \tag{B5}$$

Subtracting this result from the total value one gives Eq. (64).

c. Second-order fidelity roof

For a normalized pure state $|\phi\rangle = \alpha|\psi_+\rangle + \beta|\psi_-\rangle$, define

$$x = 2\,\mathrm{Re}(\alpha^*\beta), \qquad y = 2\,\mathrm{Im}(\alpha^*\beta), \qquad z_\phi = |\alpha|^2 - |\beta|^2. \tag{B6}$$

The spectrum of the two-qubit reduced state is

$$p_\pm = \frac{1}{3} + \frac{x}{6} \pm \frac{\sqrt{3}\,y}{6}, \quad p_3 = \frac{1}{3} - \frac{x}{3}, \tag{B7}$$

and therefore

$$\mathcal{A}_2^F(|\phi\rangle) = \mathcal{N}_2(|\phi\rangle) = \sqrt{p_+ p_-} + \sqrt{p_+ p_3} + \sqrt{p_- p_3}. \tag{B8}$$

To obtain a lower bound at fixed z_ϕ, set $z = z_\phi$ and $r = \sqrt{1 - z^2}$. The three reduced eigenvalues obey

$$\sum_i p_i = 1, \qquad \sum_i p_i^2 = \frac{3 - z^2}{6}. \tag{B9}$$

Minimizing $\sum_{i<k}\sqrt{p_i p_k}$ is equivalent to minimizing $\sum_i \sqrt{p_i}$. For $0 < r < 1$, the eigenvalues are positive, and a stationary point under the two constraints satisfies $1/(2\sqrt{p_i}) = \alpha + 2\beta p_i$. The strictly convex function on the left intersects a line in at most two positive points, so at least two eigenvalues are equal. Comparing the two resulting branches gives the minimum at

$$(p_1, p_2, p_3) = (a, a, b), \qquad a = \frac{2 + r}{6}, \quad b = \frac{1 - r}{3}, \tag{B10}$$

up to permutation. The endpoints follow by continuity (including the zero eigenvalue at $r = 1$). Hence

$$\mathcal{N}_2(\phi) \geq a + 2\sqrt{ab} = \frac{2 + r + 2\sqrt{2(2 + r)(1 - r)}}{6}. \tag{B11}$$

To compare this with $(1 + |z|)/2$, square the nonnegative sides of

$$2\sqrt{2(2 + r)(1 - r)} \;\geq\; 1 - r + 3\sqrt{1 - r^2}. \tag{B12}$$

The difference of their squares is $6(1 - r)(1 - \sqrt{1 - r^2}) \geq 0$. This proves the pointwise bound

$$\mathcal{A}_2^F(|\phi\rangle) \geq \frac{1 + |z_\phi|}{2}. \tag{B13}$$

Consequently, consider an arbitrary pure-state decomposition $\rho(\lambda) = \sum_i p_i |\phi_i\rangle\langle\phi_i|$, with $|\phi_i\rangle = \alpha_i|\psi_+\rangle + \beta_i|\psi_-\rangle$. Since this decomposition must reproduce the Bloch vector of $\rho(\lambda)$ inside the support $\mathrm{span}\{|\psi_+\rangle, |\psi_-\rangle\}$, its z-components satisfy

$$\sum_i p_i z_i = 2\lambda - 1, \qquad z_i = |\alpha_i|^2 - |\beta_i|^2. \tag{B14}$$

Using the bound above for each pure state in the decomposition, we obtain

$$\begin{aligned} \sum_i p_i \mathcal{A}_2^F(|\phi_i\rangle) &\geq \frac{1}{2} + \frac{1}{2}\sum_i p_i |z_i| \\ &\geq \frac{1}{2} + \frac{1}{2}\left|\sum_i p_i z_i\right| \\ &= \frac{1}{2} + \left|\lambda - \frac{1}{2}\right|. \end{aligned} \tag{B15}$$

This lower bound is tight. For $\lambda \geq 1/2$, write

$$\rho(\lambda) = (2\lambda - 1)|\psi_+\rangle\langle\psi_+| + 2(1 - \lambda)\rho_*, \tag{B16}$$

where

$$\rho_* = \frac{1}{2}\left(|\psi_+\rangle\langle\psi_+| + |\psi_-\rangle\langle\psi_-|\right) = \frac{1}{3}\sum_{k=0}^{2}|\phi_k\rangle\langle\phi_k| \tag{B17}$$

with

$$|\phi_k\rangle = \frac{1}{\sqrt{2}}\left(|\psi_+\rangle + e^{2\pi ik/3}|\psi_-\rangle\right). \tag{B18}$$

Since $\mathcal{A}_2^F(|\psi_+\rangle) = 1$ and $\mathcal{A}_2^F(|\phi_k\rangle) = 1/2$, this decomposition has average value λ. The case $\lambda \le 1/2$ is obtained by exchanging $|\psi_+\rangle$ and $|\psi_-\rangle$. This attains the lower bound and proves Eq. (65).

d. *Third-order values*

The three-qubit reduction has spectrum

$$\left\{\frac{\lambda}{2}, \frac{\lambda}{2}, \frac{1-\lambda}{2}, \frac{1-\lambda}{2}\right\}. \tag{B19}$$

Substituting into the total purity and fidelity functionals reproduces the total values obtained from Eqs. (66) and (67). The complementary-partition relations give

$$\mathcal{A}_3^{Q,\mathcal{P}} = \tfrac{2}{3}\mathcal{A}_1^{Q,\mathcal{P}} = \tfrac{2}{3}, \qquad \mathcal{A}_3^{Q,F} = \tfrac{1}{3}\mathcal{A}_1^{Q,F} = \tfrac{1}{3}. \tag{B20}$$

Subtracting these constant quantum values yields the classical contributions quoted in the main text.

2. Rank-two purity roofs

Let $\{|e_0\rangle, |e_1\rangle\}$ be an orthonormal basis for the support of a rank-two state. Represent the state by an effective Bloch vector $\mathbf{r}$ and pure vectors by $\mathbf{n}$, $|\mathbf{n}| = 1$. Linearity of reduction makes the purity functional quadratic:

$$f(\mathbf{n}) = a + \mathbf{b}\cdot\mathbf{n} + \mathbf{n}^{\mathsf{T}} Q \mathbf{n}, \tag{B21}$$

where Q is a real symmetric 3×3 matrix. Its convex roof on this support is

$$f^{\mathrm{CR}}(\mathbf{r}) = a + \mathbf{b}\cdot\mathbf{r} + \mathbf{r}^{\mathsf{T}} Q \mathbf{r} + \lambda_{\min}(Q)(1 - |\mathbf{r}|^2). \tag{B22}$$

Indeed, an ensemble has covariance $\Sigma = \sum_i p_i(\mathbf{n}_i - \mathbf{r})(\mathbf{n}_i - \mathbf{r})^{\mathsf{T}} \ge 0$ with $\operatorname{Tr}\Sigma = 1 - |\mathbf{r}|^2$. Thus $\operatorname{Tr}(Q\Sigma) \ge \lambda_{\min}(Q)\operatorname{Tr}\Sigma$. Equality is attained by the two endpoints where a line through $\mathbf{r}$ parallel to a minimum-eigenvalue eigenvector of Q meets the unit sphere, with weights chosen to average to $\mathbf{r}$.

For explicit evaluation, put

$$R_0 = \operatorname{Tr}_{N-t}(EE^\dagger/2), \qquad R_a = \operatorname{Tr}_{N-t}(E\sigma_a E^\dagger/2), \tag{B23}$$

where $E = (|e_0\rangle, |e_1\rangle)$ and σ_a are Pauli matrices on the effective support. With $c = (t+1)/t$ the coefficients are

$$a = c(1 - \operatorname{Tr} R_0^2), \qquad b_a = -2c\operatorname{Tr}(R_0 R_a), \tag{B24}$$

$$Q_{ab} = -c\operatorname{Tr}(R_a R_b). \tag{B25}$$

The same quadratic expression at $\mathbf{r}$ gives the total purity measure, so the classical contribution simplifies to

$$\mathcal{A}_t^{C,\mathcal{P}}(\rho) = -\lambda_{\min}(Q)(1 - |\mathbf{r}|^2). \tag{B26}$$

Here $Q \le 0$ because $-Q$ is a Gram matrix. The two factors separate the dependence on the support and on the mixing within it. Substitution gives the purity-based entries for the four rank-two states in Table I and the rank-two HOAP reductions in Fig. 2(a).

3. Proofs of the negativity saturation criteria

a. *Anticoherent support*

Proof of Theorem 1. The ordinary-negativity equality is established in Theorem 1 of Ref. [28]; it holds for every such subspace. Its convex-roof consequence also follows directly: every pure vector in every decomposition of ρ belongs to S and has the same maximal value. For the converse, any normalized vector $|v\rangle \in S$ can occur with nonzero probability in a pure-state decomposition, since $\rho - \epsilon|v\rangle\langle v| \ge 0$ for sufficiently small $\epsilon > 0$. If one such vector had value below one, this would yield an ensemble average below one. Hence every vector in S must be t-AC. The total value is then one and the classical part zero. □

b. *Orthogonal supports after partial transposition*

Let $X_i = (|\psi_i\rangle\langle\psi_i|)^{T_t}$ have pairwise orthogonal supports. Since each X_i is Hermitian, $\rho^{T_t} = \sum_i p_i X_i$ is a direct sum on these supports. Its nonzero eigenvalues are therefore the union, with multiplicities, of those of the blocks $p_i X_i$. Summing the absolute values of the negative eigenvalues gives

$$\mathcal{N}_t(\rho) = \sum_i p_i \mathcal{N}_t(|\psi_i\rangle). \tag{B27}$$

Convexity of the ordinary negativity and the definition of the convex roof then imply

$$\mathcal{N}_t(\rho) \le \mathcal{N}_t^{\mathrm{CR}}(\rho) \le \sum_i p_i \mathcal{N}_t(|\psi_i\rangle) = \mathcal{N}_t(\rho). \tag{B28}$$

Equality holds throughout, proving both saturation and optimality of the displayed decomposition. For the occurrence of this orthogonality in AC subspaces, see Appendix F of Ref. [28].

Appendix C: Numerical methods and scope of the bounds

1. Covariant-channel tests

For Hilbert–Schmidt normalized irreducible tensors on the spin-$N/2$ sector, the maps

$$\Phi_K(X) = \frac{N+1}{2K+1}\sum_{M=-K}^{K} T_{KM} X T_{KM}^\dagger, \tag{C1}$$

for $K = 0, \ldots, N$ are CPTP covariant channels. They are the vertices of the fixed-sector covariant-channel simplex: covariance makes the Choi operator scalar on each irreducible sector of $U_j \otimes \overline{U}_j$ [47, 48] [49]. This parameterization respects complete positivity, unlike independent sampling of multipole attenuation factors.

An exploratory test used 400 random density operators for each $N = 2, \ldots, 6$, each vertex in Eq. (C1), and every $1 \le t < N$. A random rank was chosen uniformly from $1, \ldots, N+1$ and a density operator formed as $GG^\dagger / \operatorname{Tr}(GG^\dagger)$ from a complex Gaussian matrix of that size. No resolved decrease of the trace-distance or fidelity total was found. An identity-channel fidelity comparison near zero eigenvalues differed by approximately 10^{-8} from zero because of floating-point square roots; trace-norm differences were at roundoff level. These finite tests only support Conjecture 1. A proof for all inputs at the vertices would extend to their convex mixtures by concavity of the total functionals.

2. States and feasible-ensemble optimization

The HOAP input vectors used in both blocks of Fig. 2, in the ordered Dicke basis $m = N/2, N/2-1, \ldots, -N/2$, are proportional to

$$v_4 = (1, 0, i\sqrt{2}, 0, 1), \tag{C2}$$
$$v_5 = (0, 1, 0, 0, 1, 0), \tag{C3}$$
$$v_6 = (0, 1, 0, 0, 0, 1, 0), \tag{C4}$$
$$v_7 = (2, 0, 0, -\sqrt{7}, 0, 0, -\sqrt{7}, 0), \tag{C5}$$
$$v_8 = (\sqrt{5}, 0, 0, 0, -\sqrt{14}, 0, 0, 0, -\sqrt{5}), \tag{C6}$$
$$v_9 = (\sqrt{13}, 0, 0, 0, \sqrt{30}, 0, 0, 0, -\sqrt{21}, 0), \tag{C7}$$
$$v_{10} = (0, \sqrt{5}, 0, 0, 0, \sqrt{6}, 0, 0, 0, -\sqrt{5}, 0). \tag{C8}$$

Each vector is divided by its norm. Their AC orders are $2, 1, 3, 2, 3, 2, 3$, respectively [40]. For a bipartition into t and $N - t$ symmetric qubits, the isometry in excitation-number coordinates is

$$V_{ab,k} = \delta_{k,a+b} \sqrt{\frac{\binom{t}{a}\binom{N-t}{b}}{\binom{N}{k}}}, \tag{C9}$$

which gives all reductions without embedding into the full 2^N space.

For a rank-r mixed state, write $\rho = BB^\dagger$ using its positive spectral square root on its support. An $m \times r$ complex isometry U, $U^\dagger U = \mathbb{1}_r$, defines subnormalized pure vectors

$$|x_i\rangle = \sum_{\alpha=1}^{r} U_{i\alpha} B_{:\alpha}, \qquad \rho = \sum_i |x_i\rangle\langle x_i|. \tag{C10}$$

The objective is $\sum_i p_i \mathcal{A}_t^{\mathcal{P}}(x_i/\sqrt{p_i})$ with $p_i = \langle x_i | x_i \rangle$. Every feasible objective is an upper bound on the convex roof.

The independent purity-roof calculations for this revision used $m = r + 2$ and a thin complex QR factorization to maintain the isometry constraint. The real and imaginary parts of an unconstrained matrix were optimized by L-BFGS-B with automatic derivatives, using eight random starts and at most 800 iterations for each higher-rank HOAP reduction. For the rank-four state in Eq. (87), 24 starts and at most 1500 iterations were used separately for $t = 1, 2$. The relative objective-change and gradient stopping thresholds were 10^{-13} and 10^{-9}, with random seed 20260930. Pure and rank-two states were evaluated analytically rather than optimized.

The saved ensembles reconstruct their target density matrices with Hilbert–Schmidt residual below 8×10^{-16}. Their objectives were checked again by explicitly normalizing each component and reducing it. For the rank-four state this initial search gave the upper bounds $q_1 \le 0.7325$ and $q_2 \le 0.6731$. These numerical estimates are superseded by the exact algebraic characterizations and global certificates in Ref. [66]; the optimal second-order ensemble has twelve components. The higher-rank HOAP estimates remain numerical feasible-ensemble bounds with the stated residual, not rigorous interval-arithmetic certificates or claims of global convergence.

For comparison, Cauchy–Schwarz applied to the Schmidt coefficients of a pure state gives $\mathcal{A}_t^{\mathcal{P}}(\psi) \ge [2\mathcal{N}_t(\psi)/t]^2$. Jensen's inequality and convexity of ordinary negativity then imply

$$\mathcal{A}_t^{\mathcal{Q},\mathcal{P}}(\rho) \ge [2\mathcal{N}_t(\rho)/t]^2. \tag{C11}$$

For Eq. (87), direct partial transposition gives $\mathcal{N}_1 = 19/60$ and $\mathcal{N}_2 = 11/30$, yielding $q_1 \ge 361/900$ and $q_2 \ge 121/900$. The complementary-order identities improve the corresponding bounds at $t = 3, 4$. These negativity bounds are much weaker than the certificates in Ref. [66], which are used for the rank-four purity allocation ($\mathcal{P}$) in Fig. 3.

Fidelity-based figures. For Fig. 2(b) and the fidelity pies (F) in Fig. 3, the objective is instead $\sum_i p_i \mathcal{A}_t^F(x_i/\sqrt{p_i})$, evaluated from the Schmidt singular values of each component. The numerical search uses the same isometry parameterization, up to 16 starts and 1800 iterations, and ensembles with $r+2$, $2r$, or r^2 components, enlarged when needed to include a saved initial ensemble. Spectral decompositions and saved purity and fidelity ensembles supply initial upper bounds. The objective-change and gradient thresholds are 2×10^{-13} and 2×10^{-9}. A search can stop early when its gap above $2\mathcal{N}_t(\rho)/t$ is below 3×10^{-9}. This ordinary-negativity value is a lower bound, not a replacement for the convex roof. The 53 saved numerical ensembles reconstruct their targets with residual below 8×10^{-16}; independent reduced-spectrum evaluations agree with their objectives within 1.1×10^{-7}. These are floating-point checks, not global or interval-arithmetic certificates. The raw data retain both bounds and identify analytic entries. Higher orders in the pies follow from Eq. (35).

Several mixed-state fidelity pies require no numerical minimization. Every pure vector in the support of Eq. (80) has first-order reduced spectrum $\{2/3, 1/3\}$, giving $\mathcal{A}_1^{Q,F} = 2\sqrt{2}/3$. Likewise, the support of Eq. (85) has constant second-order reduced spectrum $\{1/2, 1/3, 1/6\}$, so $\mathcal{A}_2^{Q,F} = (\sqrt{6}+\sqrt{3}+\sqrt{2})/6$; its first-order value follows from the orthogonal-support argument in Sec. IV A. For Eq. (84), the equal mixture of $\sqrt{1/6}\,|\psi_1\rangle \pm \sqrt{5/6}\,|\psi_2\rangle$ attains $\mathcal{A}_1^{Q,F} = 2\sqrt{2}/3$, matching the ordinary-negativity lower bound. The spin-2 values are given in Sec. IV; the MMS and pure-state endpoints follow directly from separability and the Schmidt spectra, respectively. The remaining mixed-state pies use the numerical bounds described above.

Fidelity bounds used in Table I. For the state in Eq. (84), the second-order ordinary negativity is $(6 + \sqrt{13})/12$. Its spectral ensemble has average fidelity anticoherence $(1 + \sqrt{3})/3$, giving the displayed exact lower and upper bounds on r. Neither endpoint is asserted to be the convex roof.

The certified purity roofs yield stronger lower bounds for the rank-four state than ordinary negativity. Indeed, for reduced eigenvalues λ_i of a normalized pure state, $\sqrt{\lambda_i\lambda_j} \le 1/2$ implies

$$\begin{aligned}\mathcal{A}_t^F(\psi) &= \frac{2}{t}\sum_{i<j}\sqrt{\lambda_i\lambda_j} \\ &\ge \frac{4}{t}\sum_{i<j}\lambda_i\lambda_j = \frac{2}{t+1}\mathcal{A}_t^{\mathcal{P}}(\psi). \end{aligned} \tag{C12}$$

Averaging over any pure-state ensemble and minimizing therefore gives

$$\mathcal{A}_t^{Q,F}(\rho) \ge \frac{2}{t+1}\mathcal{A}_t^{Q,\mathcal{P}}(\rho). \tag{C13}$$

For Eq. (87), this proves $f_1 \ge q_1 \approx 0.7325$ and $f_2 \ge \frac{2}{3}q_2 \approx 0.4471$. The saved feasible ensembles give the numerical upper estimates $f_1 \lesssim 0.8554$ and $f_2 \lesssim 0.4909$, also reported in Table I. Their reconstruction residuals are below 7×10^{-16}; the bounds remain numerical rather than exact certificates. The nonunit total entries in the table follow directly from the reduced spectra through Eq. (27).

[1] J. Zimba, Electronic Journal of Theoretical Physics **3**, 143 (2006).
[2] M. Rudziński, A. Burchardt, and K. Życzkowski, Quantum **8**, 1234 (2024).
[3] G. Björk, A. B. Klimov, P. de la Hoz, M. Grassl, G. Leuchs, and L. L. Sánchez-Soto, Phys. Rev. A **92**, 031801(R) (2015).
[4] J. Crann, R. Pereira, and D. W. Kribs, J. Phys. A: Math. Theor. **43**, 255307 (2010).
[5] D. Baguette, F. Damanet, O. Giraud, and J. Martin, Phys. Rev. A **92**, 052333 (2015).
[6] E. Serrano-Ensástiga and D. Braun, Phys. Rev. A **101**, 022332 (2020).
[7] D. Baguette, T. Bastin, and J. Martin, Phys. Rev. A **90**, 032314 (2014).
[8] P. Kolenderski and R. Demkowicz-Dobrzanski, Phys. Rev. A **78**, 052333 (2008).
[9] C. Chryssomalakos and H. Hernández-Coronado, Phys. Rev. A **95**, 052125 (2017).
[10] A. Z. Goldberg and D. F. V. James, Phys. Rev. A **98**, 032113 (2018).
[11] J. Martin, S. Weigert, and O. Giraud, Quantum **4**, 285 (2020).
[12] H. Ferretti, Y. B. Yilmaz, K. Bonsma-Fisher, A. Z. Goldberg, N. Lupu-Gladstein, A. O. T. Pang, L. A. Rozema, and A. M. Steinberg, Optica Quantum **2**, 91 (2024).
[13] J. M. Robbins and M. V. Berry, Journal of Physics A: Mathematical and General **27**, L435 (1994).
[14] P. Aguilar, C. Chryssomalakos, E. Guzmán-González, L. Hanotel, and E. Serrano-Ensástiga, Journal of Physics A: Mathematical and Theoretical **53**, 065301 (2020).
[15] C. Chryssomalakos, L. Hanotel, E. Guzmán-González, and E. Serrano-Ensástiga, Mod. Phys. Lett. A **37**, 2250184 (2022).
[16] J. R. Hervas, A. Z. Goldberg, A. S. Sanz, Z. Hradil, J. Řeháček, and L. L. Sánchez-Soto, Phys. Rev. Lett. **134**, 010804 (2025).
[17] S.-H. Chiew and M. Gessner, Phys. Rev. Res. **4**, 013076 (2022).
[18] S. Massar and S. Popescu, Phys. Rev. Lett. **74**, 1259 (1995).
[19] N. Gisin and S. Popescu, Phys. Rev. Lett. **83**, 432 (1999).
[20] A. Peres and P. F. Scudo, Phys. Rev. Lett. **86**, 4160 (2001).
[21] A. Peres and P. F. Scudo, Phys. Rev. Lett. **87**, 167901 (2001).
[22] A. Peres and P. Scudo, Journal of Modern Optics **49**, 1235 (2002).
[23] D. Collins and S. Popescu, arXiv preprint quant-ph/0401096 (2004), arXiv:quant-ph/0401096.
[24] E. Chitambar and G. Gour, Rev. Mod. Phys. **91**, 025001 (2019).
[25] S. D. Bartlett, T. Rudolph, and R. W. Spekkens, Rev. Mod. Phys. **79**, 555 (2007).
[26] G. Gour and R. W. Spekkens, New J. Phys. **10**, 033023 (2008).
[27] G. Gour, I. Marvian, and R. W. Spekkens, Phys. Rev. A **80**, 012307 (2009).
[28] E. Serrano-Ensástiga, C. Chryssomalakos, and J. Martin, Phys. Rev. A **111**, 022435 (2025).
[29] F. Bouchard, P. de la Hoz, G. Björk, R. W. Boyd, M. Grassl, Z. Hradil, E. Karimi, A. B. Klimov, G. Leuchs, J. Řeháček, and L. L. Sánchez-Soto, Optica **4**, 1429 (2017).
[30] J. Denis, C. Read, and J. Martin, SciPost Phys. Core **9**, 001 (2026).
[31] A. Z. Goldberg, J. R. Hervas, A. S. Sanz, A. B. Klimov, J. Řeháček, Z. Hradil, M. Hiekkamäki, M. Eriksson, R. Fickler, G. Leuchs, and L. L. Sánchez-Soto, Quantum Science and Technology **10**, 015053 (2024).

[32] J. Bringewatt, L. Zaporski, M. Radzihovsky, J. Albert, A. V. Gorshkov, V. Vuletic, and G. Bentsen, Butterfly echo protocol for axis-agnostic Heisenberg-limited metrology (2026), arXiv:2602.23332 [quant-ph].

[33] D. Baguette and J. Martin, Phys. Rev. A **96**, 032304 (2017).

[34] E. Majorana, Il Nuovo Cimento (1924-1942) **9**, 43 (1932).

[35] M. Aulbach, D. Markham, and M. Murao, New J. Phys. **12**, 073025 (2010).

[36] M. Aulbach, Int. J. Quantum Inf. **10**, 1230004 (2012).

[37] P. de la Hoz, A. B. Klimov, G. Björk, Y.-H. Kim, C. Müller, C. Marquardt, G. Leuchs, and L. L. Sánchez-Soto, Phys. Rev. A **88**, 063803 (2013).

[38] L. L. Sánchez-Soto, A. B. Klimov, P. de la Hoz, and G. Leuchs, J. Phys. B: At. Mol. Opt. Phys. **46**, 104011 (2013).

[39] D. A. Varshalovich, A. N. Moskalev, and V. K. Khersonskii, *Quantum Theory of Angular Momentum* (World Scientific, 1988).

[40] J. Denis and J. Martin, Phys. Rev. Res. **4**, 013178 (2022).

[41] O. Giraud, D. Braun, D. Baguette, T. Bastin, and J. Martin, Phys. Rev. Lett. **114**, 080401 (2015).

[42] I. Marvian and R. W. Spekkens, Nat. Commun. **5**, 3821 (2014).

[43] I. Marvian and R. W. Spekkens, Phys. Rev. A **90**, 014102 (2014).

[44] A. S. Holevo, Remarks on the classical capacity of quantum channel (2002), arXiv:quant-ph/0212025.

[45] C. Müller, Diffusive spin transport, in *Entanglement and Decoherence: Foundations and Modern Trends*, edited by A. Buchleitner, C. Viviescas, and M. Tiersch (Springer Berlin Heidelberg, Berlin, Heidelberg, 2009) pp. 277–314.

[46] Á. Rivas and A. Luis, Phys. Rev. A **88**, 052120 (2013).

[47] E. Chang, J. Kim, H. Kwak, H. H. Lee, and S.-G. Youn, Reviews in Mathematical Physics **34**, 2250021 (2022).

[48] T. Aschieri, B. Ruba, and J. P. Solovej, Communications in Mathematical Physics **405**, 298 (2024).

[49] Ł. Rudnicki, A. B. Klimov, A. Muñoz, G. Leuchs, and L. L. Sánchez-Soto, Symmetry **18**, 1146 (2026).

[50] T. Ichikawa, T. Sasaki, I. Tsutsui, and N. Yonezawa, Physical Review A **78**, 052105 (2008).

[51] L. Henderson and V. Vedral, J. Phys. A: Math. Gen. **34**, 6899 (2001).

[52] O. Giraud, P. Braun, and D. Braun, Phys. Rev. A **78**, 042112 (2008).

[53] S. L. Luo, Theor. Math. Phys. **143**, 681 (2005).

[54] G. Vidal and R. F. Werner, Phys. Rev. A **65**, 032314 (2002).

[55] S. Lee, D. P. Chi, S. D. Oh, and J. Kim, Phys. Rev. A **68**, 062304 (2003).

[56] M. Koashi and A. Winter, Phys. Rev. A **69**, 022309 (2004).

[57] A. Z. Goldberg, A. B. Klimov, H. deGuise, G. Leuchs, G. S. Agarwal, and L. L. Sánchez-Soto, Optics Letters **47**, 477 (2022).

[58] A. Z. Goldberg, P. de la Hoz, G. Björk, A. B. Klimov, M. Grassl, G. Leuchs, and L. L. Sánchez-Soto, Advances in Optics and Photonics **13**, 1 (2021).

[59] M. A. Nielsen, Phys. Rev. Lett. **83**, 436 (1999).

[60] R. Pereira and C. Paul-Paddock, J. Math. Phys. **58**, 062107 (2017).

[61] J. A. Gross, Phys. Rev. Lett. **127**, 010504 (2021).

[62] S. Lim, J. Liu, and A. Ardavan, Phys. Rev. A **108**, 062403 (2023).

[63] G. Vidal, Journal of Modern Optics **47**, 355 (2000).

[64] A. Z. Goldberg, Optica Quantum **2**, 14 (2024).

[65] W. Kłobus, A. Burchardt, A. Kołodziejski, M. Pandit, T. Vértesi, K. Życzkowski, and W. Laskowski, Phys. Rev. A **100**, 032112 (2019).

[66] J. Denis, T. Lacaille, J. Martin, and E. Serrano-Ensástiga, Exact purity and convex-roof certificates for spin 5/2, GitHub repository (2026), version 1.1.0, commit 455931c; proofs, exact coefficient data, and verification programs.